

Master Thesis

Consumer Perceptions of Branded Resale Channels: The Role of Design Features in Balancing Sustainability and Value

by:

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Abstract

Branded resale, where a company sells its own used products alongside new ones, is widely framed as a low-risk sustainability signal that does not eat into new product sales. This thesis tests that assumption using nearly 50,000 unsolicited Trustpilot reviews from 48 fashion and outdoor brands, comparing customer perceptions before and after a resale launch. The available comparison brands differ systematically from the brands that launched resale, which makes a standard two-way fixed effects design unreliable. Synthetic difference-in-differences reweights the control group to match each treatment brand's pre-launch trajectory and is used as the primary estimator throughout. The correction matters: a positive, non-significant effect on value sentiment under the standard approach becomes negative and significant once the comparison group is properly weighted. Across nineteen treatment brands, resale launches are associated with a decline in customer perception on average, most visibly in how customers write about price and value, with considerable variation by brand segment. A second study tests whether four design choices, integration level, messaging intensity, discount depth, and service offerings, explain that variation. None of them do, at least not individually. With only nineteen brands, the sample did not have enough power to detect anything but a large effect. One pattern survives: structural integration and sustainability messaging behave as substitutes rather than complements in shaping sustainability perception, consistent with a signalling backfire once a brand's resale design already communicates commitment. These findings complicate the optimistic narrative around branded resale and point to method choice, not just data, as central to what a study of this kind can responsibly claim.

Index

1. Introduction *5*

1.1 Background 5

1.2 Research Questions 6

2. Literature review *8*

2.1 Secondary Markets and Cannibalization 8

2.2 Sustainability Signaling and Consumer Perception 10

2.3 Theoretical Frameworks Underlying Design Choices 12

3. Conceptual Framework *13*

4. Methodology *15*

4.1 Data Collection & Sources 15

4.2 Measures 17

4.2.1 Dependent Variables 17

4.2.2 Moderators 19

4.2.3 Controls 21

4.3 Analysis 21

4.3.1 Preliminary analysis: LDA topic modelling 21

4.3.2 Study 1: Difference-in-differences 22

4.3.3 Study 2: Moderation analysis 24

5. Results *26*

5.1 Descriptive Statistics 26

5.2 Study 1 Results 30

5.3 Study 2 Results 38

6. Discussion *41*

6.1 Addressing 1st Research Question 41

6.2 Addressing 2nd Research Question 42

7. Limitations *44*

8. Practical Implications *46*

9. Conclusion *47*

10. References *49*

A. Appendix *53*

A.1 Control Brand Overview 53

A.2 Sustainability Dictionary 55

A.3 Treatment Brand Moderating Coding 56

A.4 LDA Topic Model: Technical Details 58
A.5 Sustainability Language Mention 59
A.6 Full Two-Way Fixed-Effects DiD Estimates 61
A.7 Robustness Check 61
A.8 Sensitivity Check: Moderation Analysis Using Raw Change Scores 62

1. Introduction

1.1 Background

The fashion industry has witnessed a change: brands are increasingly choosing to control their own resale channels rather than relying solely on third-party platforms like The RealReal, Vestiaire Collective, or ThredUp (Bain, 2022). For example, in October 2021, the luxury brand Sandro Paris launched “Sandro Seconde Main” (Douglass, 2022), integrating pre-owned items directly into its online platform often priced 40-60% below retail depending on its condition. Ba&sh followed with "Ba&sh Second hand" or “pre-loved” in November 2022 (FashionNetwork, 2024), Lacoste introduced its resale program in December 2024 (Luxus Plus, 2024), among many other brands. Others, like "Worn Wear" program launched by Patagonia has operated since 2013 and is frequently cited as proof that branded resale can coexist with or even strengthen new product sales.

Through their own resale channels, brands stay in control of how they present themselves, what price they set, and who their customers are by building their own resale infrastructure. The market opportunity driving this shift seems to be substantial. The World Economic Forum (WEF) estimates that circular business models could generate $4.5 trillion in economic benefits by 2030 (Ellen MacArthur Foundation & Kantar, 2024). Moreover, ThredUp's 2023 Resale Report projects that the secondhand apparel market will reach $350 billion globally by 2027, nearly doubling from $177 billion in 2022 (ThredUP, 2023). The motives behind starting first-party resale initiatives vary considerably across businesses and stakeholders. As Mara Hoffman's founder observed, while the interest in sustainability is growing, motivations range from genuine circularity commitments to commercial revenue strategies (Cope, 2022). For companies, own resale channels may thus be both a defensive move (as it preserves the brand equity in secondary markets) and an offensive opportunity (because it acquires circular economy value while showing a commitment to sustainability).

However, the more pressing tension manifests itself in consumers' thoughts, frequently long before it appears in sales figures. A brand runs the danger of upsetting consumers' perceptions of its main offering when it simultaneously sells new products and less expensive used

alternatives, namely whether the resale channel is a sincere commitment to sustainability or merely a margin-protection strategy, and whether full-price items still feel worthwhile. These changes in perception are not coincidental; rather, they are a pathway for the manifestation of cannibalization risk. Behavioral change usually follows when consumers begin to doubt the benefit of purchasing new products or lose faith in the brand's sustainability narrative. The downstream financial effects of secondary markets on demand and profitability have been shown in earlier research (Bulow, 1982, 1986; Waldman, 1997; Yin et al., 2010). However, the factors that influence these effects, specifically the consumer perceptions that precede them, have not been thoroughly examined.

Also, brands use different strategies in their resale channel designs, differentiating substantially between them. For example, integration approaches range from Patagonia's side-by-side display of new and used products (Martinko, 2020) to Sandro's separate "Seconde Main" website that only shows second-hand clothes (Sandro, n.d.). Sustainability messaging expands from prominent circular economy language ("Close the loop") to minimal framing ("Shop pre-owned"). Discount depths also vary from 30–40% (Arc'teryx, Canada Goose) to 50–70% (Armedangels, Sandro), and service offerings differ between brands that extend warranties to resale purchases and those that sell used items as they are (based on brand resale channel websites; see Table A.3).

1.2 Research Questions

The critical managerial question is not whether to launch resale, but how to design resale channels that maximize sustainability benefits while minimizing cannibalization risks. Prior research on consumer responses to sustainable initiatives has relied predominantly on survey-based measures of stated preferences and purchase intentions (Chen, 2010; Chen & Chang, 2013), which are known to overstate actual environmental concern due to social desirability bias (Lyon & Maxwell, 2011). Until now, no research has used observational data from actual market situations to investigate how particular resale channel design aspects influence consumer views. By examining unsolicited customer reviews, which capture revealed rather

than stated perceptions and thus provide a more ecologically valid assessment of how customers truly react to branded resale programs, this study fills that gap.

In theory, the most concrete proof of customer reactions would come from comparing sales data before and after reselling program begins. However, academic study involving several companies cannot obtain brand-level sales data since it is confidential. A theoretically sound substitute is consumer review data, which documents opinions as they happen rather than as they are formed in a survey environment. Reviews are unsolicited, timestamped, and publicly expressed. Previous studies have demonstrated that NLP-analyzed online customer evaluations are reliable indicators of brand perception, capturing consumer sentiment with more ecological validity than conventional survey methods (Raza et al., 2025). To better understand how branded resale channels affect consumers' views of brand sustainability and new product value, as well as how certain design elements influence these effects, this study examines consumer review data. The results will help brands make judgments about their resale strategies while also advancing theoretical knowledge of cannibalization theory and the dynamics of the circular economy.

The central question guiding this research is how branded resale channels influence consumer perceptions of brand sustainability and new product value, and how resale channel design characteristics moderate these effects. Two complementary sub-questions structure this inquiry, each addressed through a separate study. The first establishes what actually happens when brands launch resale: whether consumer perceptions shift, and in which direction. Do sustainability perceptions improve? Do concerns about durability or value emerge alongside them? And can positive and negative effects occur at the same time? This is addressed through a before-and-after analysis comparing perceptions prior to and following a resale channel launch.

The second issue relates to the circumstances in which resale is beneficial or unfavorable. It specifically explores whether strong sustainability messaging may counteract the concerns that high integration tends to raise, as well as how integration level and sustainability messaging intensity determine the amount and direction of perceptual impacts. These two design elements are given priority because they are the most accessible levers for brand

managers and are most closely related to the theoretical mechanisms at the center of this study, namely signaling theory and cannibalization risk. When there is enough variety in the data, pricing strategy and service offers are investigated as secondary moderators. Cross-sectional comparisons between brands with well-established resale networks are used to address this.

The two questions are complementary: the first captures what shifts in consumer perception when resale is introduced, and the second one identifies which design choices determine whether those shifts work in the brand's favor.

2. Literature review

The majority of studies on resale platform competition, cannibalization, and secondary markets have been conducted from a supply-side or economic modeling perspective, concentrating on the effects of secondary markets on demand volumes, prices, and margins. The consumer-side topic of what happens to a customer when the brand they purchased launches its own resale channel has never been addressed in a methodical manner. This is the main contribution of the current study, which lies at the intersection of two different streams: the literature on sustainability signaling in consumer markets and the literature on secondary market impacts on brand performance.

2.1 Secondary Markets and Cannibalization

The economic literature establishes a clear baseline concern about secondhand markets specifically. Foundational work by Bulow (1982, 1986) and Waldman (1997) demonstrates that the existence of secondary markets for durable goods reduces willingness to pay for new products, as consumers anticipate the future availability of cheaper used substitutes. Yin et al. (2010) extend this to show that used goods markets harm retailer profitability and compress brand margin on new products. Empirically, Ghose, Smith, and Telang (2006) provide direct evidence of this mechanism in action, finding that used product availability on internet exchanges measurably reduced demand for new items, with cannibalization rates varying significantly by product category and price differential. More recently, Pang et al.

(2024) examine consumer-to-consumer resale platforms and find that the net effect on primary demand depends heavily on consumer utility interdependence and repeat purchase behavior, with improvement effects sometimes outweighing cannibalization, but only under specific conditions of low commission rates and moderate new-product value.

The emergence of third-party resale platforms added further complexity: Eisenmann et al. (2006) and Parker and Van Alstyne (2005) show that intermediary platforms like ThredUp, The RealReal, and Vestiaire Collective generate competitive pressure through network effects, creating secondary markets that brands could neither control nor shape. Together, this work establishes that secondhand market cannibalization is real and documented, but context-dependent rather than inevitable, and that the governance structure of the resale channel matters.

The current body of research, whether it focuses on third-party platform competition, cannibalization dynamics, or macroeconomic secondary market effects, consistently looks at scenarios in which the brand has no control over resale, and the analysis itself goes beyond prices, revenues, and demand volumes. The literature has not addressed first-party resale as a separate event. First-party resale is defined here as the brand directly owning and running its own resale channel. More significantly, no prior study has made research on the psychology of consumers when they come across a resale channel. There is significant theoretical weight to this divergence. Channel ownership changes the strategic calculus considerably: brands that run their own resale programs can shape pricing, presentation, messaging, and service in ways that third-party platforms are structurally unable to replicate. Research on direct channel entry provides empirical grounding for this distinction. Van Crombrugge et al. (2025) show that when suppliers introduce their own direct channels, downstream partners respond strategically in ways that do not occur with third-party arrangements. Van Crombrugge et al. (2024) further demonstrate that brand-owned channels generate consumer awareness and purchase spillovers into independent channels, an effect that arises precisely because the brand's own presence carries a different signal than one mediated by an external party. This signal difference becomes especially consequential at the consumer level. When a brand facilitates resale directly, the psychological encounter differs from one mediated by an independent platform. Bai et al. (2026) show experimentally that

purchasing secondhand from an official brand retailer activates abstract, identity-driven thinking, while purchasing from an unofficial third-party retailer shifts attention toward concrete product details and reduces brand image perceptions. A consumer browsing Arc'teryx ReGear therefore occupies a meaningfully different psychological position than one encountering the same jacket on The RealReal. Whether first-party resale ultimately reduces cannibalization risk by framing used goods as brand-endorsed sustainable options, or heightens it by making price differences more salient, remains an open empirical question, and one that existing research has not addressed, because the focus has consistently fallen on demand volumes and prices rather than the consumer perceptions that precede them.

2.2 Sustainability Signaling and Consumer Perception

This study adds to the second stream, which focuses on how companies convey their environmental commitment and whether or not customers believe such signals. Branded resale is a sustainability promise that is different from traditional green marketing. Whether customers see it as a true representation of environmental concern or as a kind of marketing theater will have a significant impact on how it is perceived. Environmental statements may boost brand equity, but only if consumers believe they are genuine, as the literature on green marketing has long demonstrated. Lyon and Maxwell (2011) identify selective disclosure as the mechanism by which this fails: firms communicate positive environmental performance while concealing negative information, and consumers develop generalized skepticism even when they cannot identify specific deceptions. This skepticism is particularly consequential in fashion, where sustainability claims are widespread and rarely auditable. Torelli et al. (2012) show that when CSR (Corporate Social Responsibility) initiatives conflict with a brand's established identity, they can trigger incongruence effects that reduce rather than improve brand evaluations, suggesting that sustainability signals are not uniformly positive even when well-intentioned. In the circular economy context specifically, Camacho-Otero et al. (2018) find that consumer responses to circular initiatives depend heavily on how the initiative is framed and presented, not just on its existence, meaning design and communication choices determine whether sustainability signals land as credible or performative. Chen and Chang (2013) further show that claims lacking verifiable proof

trigger confusion and perceived risk, reducing trust rather than building it. An additional methodological limitation compounds this problem: stated preference methods consistently overstate environmental concern due to social desirability bias (Lyon & Maxwell, 2011), meaning that what consumers say they value and what they actually respond to in real markets diverge substantially. Camacho-Otero et al.'s (2018) systematic review confirms that most existing evidence is based on hypothetical scenarios and self-reported intentions, with very little observational evidence of how consumers actually behave in real market conditions.

Branded resale differs from traditional green marketing in that it is difficult to falsify. Sustainability commitments linked to future emissions goals or supply chain improvements are infamously difficult to validate, but managing a resale channel requires setting up actual infrastructure, including a functional platform, logistics, pricing control, and customer support. Also, by potentially discouraging its own customers from making full-price purchases, the brand is assuming real financial risk. In line with Spence's (1973) assertion that only expensive, challenging to replicate signals gain credibility, this willingness to bear a genuine expense is what gives the signal weight. A company that takes on the risk of cannibalization to promote resale is making a much more powerful message than one that makes a sustainability commitment with no stakes.

However, branded resale is not treated as clearly favorable in this analysis. The same characteristics that provide credibility to the signal might also work against it. Customers may begin comparing costs in ways that subtly undermine the perceived value of purchasing new goods when new and used goods are housed under the same brand. The signal works both ways: resale may strengthen how customers view the brand's sustainability commitment while at the same time raising doubts about whether new items are worth the price. Which impact predominates will depend on the channel's actual design. This study differs from previous research in that it captures this tension using unsolicited, real-world assessments rather than survey responses.

2.3 Theoretical Frameworks Underlying Design Choices

When two versions of the same product sit side by side on a page, something shifts in how shoppers process what they're looking at. Trope and Liberman (2010) call this a collapse of psychological distance: things that feel remote get handled abstractly, in terms of brand values and identity; things that feel close trigger concrete, detail-focused thinking. Patagonia puts used and new jackets on the same product page. The shopper who arrived thinking about sustainability ends up comparing prices instead. This is why the integration level is the first variable examined. Placing new and used products effectively directs the consumer attention on price, almost by structural design rather than explicit intent.

Petty and Cacioppo (1986) make a related point about how attitudes actually form. There are two routes: one involves genuinely engaging with an argument, the other relies on shortcuts. Which one a person takes depends on how motivated they are to pay attention. Sustainability messaging that is vague or low-effort rarely motivates the first route, so shoppers fall back on whatever signal is most visible. In a resale setting, that is usually price. For this reason, messaging intensity is important to assess, strong sustainability framing gives customers a reason to take the deeper route rather than just reaching for the price comparison. Without sustainability framing, price typically prevails as the primary decision driver.

Price is stickier than brands tend to assume. Mayhew and Winer (1992) show that consumers carry internal benchmarks and update them based on what they repeatedly see. Show the same jacket at 40% off often enough and the full-price version starts to feel hard to justify, even when nothing about the product itself has changed. Discount depth is the third variable for exactly that reason. The more often customers see the resale price, the more that number starts to feel like the real one, and the full-price item starts to feel like it needs justifying.

The last design is about service offerings. The researchers Vargo and Lusch (2014) argue that value is not something contained in a product but built through the experience surrounding it. Repair programs, trade-in schemes, and extended warranties integrate circularity more seriously. They are evidence that a brand believes in what it sells enough to stand behind it after the sale. A resale channel that includes this kind of infrastructure sends

a meaningfully different signal than one that only moves inventory. And unlike messaging, you cannot fake a repair programme the way you can inflate a sustainability landing page. That operational commitment is harder to dismiss, which is why service offerings are the fourth variable and the reason why they are expected to support both sustainability and new-product value without raising the scepticism that heavy messaging sometimes does.

How each of these designs play out in practice, and whether integration and messaging interact in the way construal level theory suggests, is what second research question addresses.

3. Conceptual Framework

Branded resale influence customer perception in two different ways. On the one hand, it can improve a company's reputation for sustainability. Adopting a resale channel requires significant infrastructure, logistics, and continuous operating spending, making it a costly decision. This sets it apart from the environmental messaging that customers are used to ignoring. According to Spence's (1973) signaling theory, a credible signal must be expensive and difficult to duplicate; a successful resale business may satisfy these requirements in ways that a marketing effort does not. Customers may feel more confident in the brand's commitment to the environment after realizing this. However, comparing new and used versions of the same product at drastically different price points might raise questions. Customers may begin to question if the new product is truly worth the price or whether the reduced resale price suggests anything about the product's long-term durability. The cannibalization mechanism that the economics research has long documented is activated by this type of direct comparison, which increases price sensitivity and directs attention toward value differences: a cheaper choice in plain sight tends to draw down the perceived worth of the full-price option. Which of these two effects wins out, or whether both operate at the same time, depends on design. Two dimensions matter most here. The first is integration level, meaning how visibly new and used products are presented together. High integration makes the price gap hard to miss, and while that visibility also signals sustainability commitment, it tends to tip consumer attention toward price comparison rather than brand

values. The second is sustainability messaging intensity, meaning how explicitly the brand frames resale as an environmental act. Prominent, well-constructed messaging may counteract the value concerns that high integration triggers by giving consumers a different interpretive lens, one focused on lifecycle responsibility rather than price comparison. These two dimensions and the way they interact form the primary moderating focus of this study. Discount depth and service offerings are included as secondary moderators given their theoretical relevance, but are only examined where the data provide enough variation to do so meaningfully.

The theoretical frameworks generate fairly specific predictions for each dimension. What each of these dimensions is expected to do, and why, becomes more clear when the relevant theoretical frameworks are considered in integration level cuts in two directions. When used products appear alongside new ones, consumers tend to read that as a genuine sustainability commitment only because the juxtaposition is visible, and the reputational cost of doing it badly is real. But that same proximity makes the price gap hard to miss, which pulls down how consumers value the new product. Construal level theory predicts both effects: physical closeness sharpens price comparison. Messaging intensity then determines whether brands can hold that tension together. Strong, explicit sustainability framing gives consumers something to weigh besides price, softening the downward pull on product value that high integration would otherwise produce. Strip the framing away or make it vague and price takes over. This interaction between the two variables is the study's central prediction. Discount depth works differently. Steep resale discounts erode consumers' internal price benchmarks over time, making the full-price item progressively harder to justify without doing much for sustainability perception on their own. Service offerings are a different category entirely. Repair programs, warranties, and trade-in schemes require real operational investment and resist easy imitation, which makes them a more credible sustainability signal than the other dimensions. They are expected to lift sustainability perception without hurting how consumers value the new product. A brand that services what it sells implies the product was worth buying.

4. Methodology

4.1 Data Collection & Sources

Trustpilot is the primary data source[1]: an independent review platform where customers post unsolicited ratings and comments about brand experiences.

The dataset covers 49,736 reviews from 48 brands in fashion and outdoor wear, collected between March 2011 and April 2026. Most observations are between 2018 and 2026, and the pre-2015 period has too few reviews per brand to work as a reliable baseline. The treatment group is 21 brands running their own resale channels, with 26,856 reviews in total: 5,457 submitted before each brand's launch date and 21,399 after. The control group is 27 brands in comparable product categories with no confirmed resale programme, covering 22,880 reviews. Together, treatment and control brands have three segments: contemporary fashion, outdoor and athletic, and premium fashion.

Rating distributions in both groups show a bimodal pattern, with most reviews sitting at one star or five stars and relatively few in between. This fits the J-shaped distribution that Hu, Zhang, and Pavlou (2009) document for online reviews: customers who had a terrible or exceptional experience are much more likely to bother writing a review than those who felt somewhere in the middle.

Treatment brands were identified through two major resale infrastructure providers: Trove, which works primarily with North American brands, and Faume, which operates primarily in European markets. Both platforms publish public lists of their partners, which makes the treatment group selection transparent and replicable. Control brands were selected within the same market segment as their treatment counterparts, in which each operates in a comparable product category but has no confirmed first-party resale program. Even though control brands

[1] Avis Vérifiés and Bazaarvoice were considered as alternative data sources. Both cover European fashion labels adequately, but Avis Vérifiés restricts programmatic access and Bazaarvoice data is not organised in a way that allows tracking brands over time. The Wayback Machine was explored as a way to retrieve historical review data, but Trustpilot pages are not archived there in any usable form. Reddit was also considered given that its discussions tend to reflect more balanced consumer opinions than review platforms, but the application for API access was denied.

are matched to treatment brands at the segment level (Section 4.3.2), they are not identical in every respect. Control brand customers tend to write more positive reviews on average and show sustainability language more often, particularly for eco-positioned niche brands such as Organic Basics and Rab. Section 5.1 documents these differences directly; they motivate the reliance on synthetic difference-in-differences rather than a simple comparison of these raw groups.

Not all 21 treatment brands have enough pre-launch observations for the difference-in-differences design to work. Arc'teryx has no pre-launch reviews and Patagonia only 2, because both resale programs predate meaningful Trustpilot coverage. These two are excluded from Study 1. Three further brands fall below the 50-observation threshold on one side of the cutoff: Balzac Paris has only 32 pre-launch reviews, The North Face 39, and Claudie Pierlot only 40 post-launch reviews. These are borderline cases and their results are interpreted with caution. The remaining 16 treatment brands clear the threshold on both sides. Among the control brands, Jottnar and Vuori each have only 99 reviews and are flagged as low volume, with their inclusion tested in robustness checks. Review volume varies considerably across the sample, and brands with larger review counts will probably carry more weight in the regression estimates, for this reason, standard errors are clustered at the brand level to account for the difference. Lacoste raises a separate concern: it has only 91 post-launch reviews against 557 pre-launch observations. That imbalance may undermine the reliability of its pre-post comparison, so its results are treated with caution. Table 1 below presents the sample reconciliation for all 21 treatment brands. A full overview of the 27 control brands, including segment classification and review counts, is provided in Appendix A.1.

Table 1. Treatment Brand Sample Reconciliation

Brand	Segment	Launch	Pre n	Post n	Study 1	Study 2
Ba&sh	CF	Nov 2022	65	568	✓	✓
Sandro	CF	Oct 2021	116	235	✓	✓
Maje	CF	Oct 2022	106	188	✓	✓

Brand	Segment	Launch	Pre n	Post n	Study 1	Study 2
Claudie Pierlot	CF	Feb 2024	70	40	post < 50	✓
Balzac Paris	CF	Nov 2021	32	1,448	pre < 50	✓
Soeur	CF	Feb 2025	284	298	✓	✓
Steve Madden	CF	Jan 2022	92	1,145	✓	✓
Armedangels	CF	Mar 2022	479	1,011	✓	✓
Studio Anneloes	CF	Feb 2024	197	3,208	✓	✓
Levi's	CF	Oct 2020	288	586	✓	✓
Hugo Boss	PF	Apr 2022	366	508	✓	✓
Lacoste	PF	Dec 2024	557	91	✓	✓
Canada Goose	PF	Jan 2023	434	2,047	✓	✓
Michael Kors	PF	Aug 2022	679	412	✓	✓
Lululemon	OA	Apr 2022	611	595	✓	✓
Allbirds	OA	Feb 2022	549	1,451	✓	✓
Aigle	OA	Oct 2020	66	953	✓	✓
Carhartt	OA	Mar 2023	425	241	✓	✓
The North Face	OA	Jun 2018	39	2,008	pre < 50	✓
Arc'teryx	OA	May 2019	0	4,000	no pre	—
Patagonia	OA	2013	2	366	no pre	—

Note: Pre n = reviews before launch date; Post n = reviews after launch date. CF = Contemporary Fashion; OA = Outdoor & Athletic; PF = Premium Fashion. Arc'teryx and Patagonia are excluded from both studies because their resale programmes predate meaningful Trustpilot review coverage.

4.2 Measures

4.2.1 Dependent Variables

Three outcome variables are measured, each capturing a different dimension of consumer perception. The first is the Trustpilot star rating, a five-point scale from one (very poor) to five (excellent). A star rating collapses the whole brand experience into a single number and does not isolate resale reactions specifically, but that limitation is less of a problem than it might seem. If a resale programme shifts how customers perceive a brand overall, the effect will show up in the rating trend whether or not customers name the resale channel. Ratings are also publicly available, timestamped, and submitted without prompting, unlike survey-based measures which are subject to the social desirability bias discussed in Section 2.2.

The second and third outcomes come from the review text itself. The first is a sustainability language score, built by counting dictionary matches across each review's combined title and body (full dictionary in Appendix A.2). The positive dictionary has 38 stems and patterns drawn from the circular economy literature (Bocken et al., 2016) and the sustainability communication framework of Delmas and Burbano (2011). These include terms relating to environmental practice (sustain*, recycl*, circular, carbon, environmental, responsible, ethical, conscious, organic, biodegradable, zero-waste, low-impact, fair-trade), circular product flows (resale, second-hand, secondhand, pre-owned, pre-loved, upcycl*, repair, reuse, renew*), product longevity claims (durable*, durability, longevity, lifecycle), and general environmental framing (green, eco, climate, planet, waste). The negative dictionary comprises seven stems targeting greenwashing scepticism (greenwash*, hypocri*, performativ*, virtue-signal*, mislead*, deceptive, and fake-green).

The net score is positive matches minus negative matches. A positive value means the review contained more positive sustainability language than negative. Otherwise, a negative value means the reviewer expressed doubt about the brand's environmental claims. Zero means no sustainability terms appeared. That last category matters more than it might seem: the measure captures what reviewers write, not what they think. A customer who cares deeply about sustainability but focuses their review on delivery or fit scores zero just the same as one who never thought about it.

The second text-based outcome is a value sentiment score, calculated using the VADER algorithm (Hutto and Gilbert, 2014). VADER was built for short social media text and outputs a compound score between -1 and 1, where -1 is maximally negative and 1 is maximally positive. Rather than running VADER on the full review text, the scoring first pulls out sentences containing value related keywords: worth, value, price, expensive, cheap, overpriced, quality, cost, affordable, bargain, deal, money, paid, pay, and spend. This follows Zeithaml's (1988) definition of perceived value as product utility relative to what is given up to obtain it, and Hutto and Gilbert's (2014) recommendation to score targeted text rather than pool everything together. Where no keyword sentence appears, the first 400 characters of the review are used as a fallback. Testing cutoffs from 200 to 600 characters produced near-identical results ($r > 0.92$), so the exact number does not affect the findings. The score shows whether price and quality language in the review is positive or negative. That makes it a proxy for perceived new-product value without needing the reviewer to mention resale at all. A lexicon-based approach was preferred over transformer-based alternatives because every step is reproducible and the results are straightforward to interpret across a corpus of 49,736 reviews.

4.2.2 Moderators

Following Neuendorf (2002), four brand-level moderator variables were coded by the author from primary sources, including each brand's resale platform, press releases, and marketing materials published around the time of programme launch, using a pre-specified codebook (see Appendix A.3). All coding was completed before the outcome data was examined.

Integration level is a five-point measure of how embedded the resale channel is within the brand's main retail environment, meaning, how visible is the resale channel towards the main brand. At 1, the resale site is a completely separate domain with no link from the main site. At 5, new and used items are side by side on the same product pages. The intermediate points are: 2 (subdomain or microsite linked from the footer only), 3 (dedicated section reachable from main navigation), and 4 (resale tab on product pages alongside new items). Patagonia's Worn Wear plugin, which displays used and new versions of the same product on one page,

is the clearest example of a 5. Most French contemporary fashion brands score 2 meaning, they have fully separate ites and minimal visual connection to the main brand.

Messaging intensity measures how prominently a brand frames its resale programme around sustainability and circularity rather than commercial convenience or price. Brands are coded on a five-point scale: 1 (no sustainability framing, purely commercial), 2 (minimal mention, sustainability present but peripheral), 3 (moderate, sustainability secondary to price or convenience framing), 4 (strong narrative, circularity is a central theme alongside commercial positioning), and 5 (core identity, the brand's sustainability commitments and its resale programme are one and the same thing). Sandro, Maje, and Ba&sh score 1; Patagonia, Allbirds, and Armedangels score 5.

Discount depth is the average percentage discount of resale prices against comparable new retail prices, consistent with price-comparison approaches used in studies of used goods markets (Yin et al., 2010; Ghose et al., 2006). For each brand, ten representative resale listings were matched to their current-season retail equivalents; the discount for each pair was calculated as ((retail price − resale price) / retail price) × 100, then averaged.[2] Michael Kors and Steve Madden both use peer-to-peer platforms where individual sellers set their own prices rather than the brand. The figures recorded for those two brands reflect what various sellers happened to charge, not a consistent brand pricing decision. This is worth keeping in mind when interpreting their results.

Service offerings counts how many additional circular economy services a brand offers beyond basic resale, out of three possibilities: repair or refurbishment, extended warranty on resale items, and trade-in or buyback. Brands with only basic buy-and-sell functionality score 1, those with one additional service score 2, and those with two or more score 3. The variable is a count rather than a scale because the three services are too different from each other to

[2]The standard deviation across the ten listings ranged from 4.8 to 22.8 percentage points, indicating moderate within-brand price dispersion. As a robustness check, the mean discount based on five listings was compared to the mean based on ten. The differences ranged from -8.9 to +3.7 percentage points (mean absolute difference: 2.7 pp), suggesting the five-listing measure is reasonably stable. The final discount depth values reported in Appendix A.2 are based on the ten-listing means

rank meaningfully. A brand that offers repairs is not obviously doing more or less than one that offers trade-ins.

4.2.3 Controls

The Study 2 model includes two segment dummies, one for Outdoor & Athletic and another one for Premium Fashion with Contemporary Fashion as the reference category. Contemporary Fashion accounts for 10 of the 19 brands and is the largest segment in the treatment group, for this reason it is the reference category. The dummies are included to absorb segment-level differences in baseline ratings and review composition that are independent of resale programme design.

4.3 Analysis

The analysis has two parts. Study 1 asks whether branded resale programmes affect consumer perceptions, and Study 2 asks which programme design features condition that effect. Even where the average effect is small, brands vary considerably in how much their perceptions change after launch, and Study 2 asks whether design characteristics explain that variation. Before doing any of the studies, an exploratory analysis was run to look at the review distributions, also at the temporal structure of the data, the baseline differences between groups, and the thematic structure of the corpus via Latent Dirichlet Allocation topic modelling.

4.3.1 Preliminary analysis: LDA topic modelling

Before applying dictionary-based scoring, an LDA topic model was estimated on all reviews containing five or more words to check whether the review contains sustainability and value language before assuming the outcome measures can detect it (Tirunillai & Tellis, 2014). Review titles and bodies were combined, lowercased, stripped of punctuation and numerals, and fed into a bag-of-words vectoriser covering 3,000 unigrams and bigrams, keeping only terms that appear in at least 30 reviews but no more than 60% of them. Seven topics gave the clearest result after testing solutions between five and ten (full model selection details in Appendix A.4). The model was estimated using scikit-learn's online variational Bayes

implementation with 30 iterations and a learning offset of 50 (Pedregosa et al., 2011). Each review was then assigned to its dominant topic for group-level comparisons. No standard errors are reported for the topic proportions because LDA is used here descriptively rather than for causal inference. This step was done to make sure the dictionary-based scores are actually picking up on something real in the reviews, rather than measuring content that consumers rarely write about.

4.3.2 Study 1: Difference-in-differences

Study 1 uses a difference-in-differences design. The control brands are what make the comparison meaningful because without them, you could not state whether a change in how people write reviews in the platform, a shift in how people talk about sustainability or general economic conditions. There would be no way to separate that from any actual effect of the resale programme.

Control brands were drawn from the same market segment as each treatment brand, forming the donor pool from which the synthetic DiD estimator constructs weighted counterfactuals. Where enough candidates existed on Trustpilot within a segment, a single treatment brand could draw on multiple control brands, which is why the control group is larger than the treatment group. A broader donor pool gives the estimator more flexibility to match each treatment brand's pre-launch trajectory. Propensity-score matching was not used because with only 21 treatment brands, one-to-one matching would leave too few observations. The resulting size imbalance between the two groups is acknowledged as a limitation and tested through the parallel trends check.

As an additional check on the reliability of the estimates, brands with less than 150 total reviews were flagged as marginal; their inclusion was then tested in robustness checks. DiD eligibility requires a treatment brand to have at least 50 reviews on each side of its launch date in the dataset. Brands not meeting this threshold are excluded or flagged as borderline, as detailed in Section 4.1. The DiD design rests on one core assumption: that without the resale launch, ratings of both groups would have moved in roughly the same direction over time. To check this, the pre-launch rating trends are compared, and a test is run to see whether

the two groups were already drifting apart before any resale programme started. If they were, the design's logic breaks down.

As a starting point, a standard two-way fixed effects DiD was estimated at the review level, with brand and quarter fixed effects and standard errors clustered at the brand level (Bertrand, Duflo, & Mullainathan, 2004). Because brands launched their resale programmes at different points in time, this was supplemented with the Callaway and Sant'Anna (2021) estimator, which is more robust to timing variations, comparing each cohort (or group) of brands that launched in the same quarter only to brands that never launched at all. Section 5 reports how these two estimates compare to each other and to the parallel trends test above, and explains why the synthetic difference-in-differences approach below is treated as the main specification for Study 1.

Study 1's main analysis uses the synthetic difference-in-differences (SDID) estimator from Arkhangelsky et al. (2021). Reviews are first aggregated to brand-quarter means for each outcome, producing a panel of brand b observed in quarter t. For each treated brand, a synthetic control is built from the 27 control brands. Intuitively, this works like an ordinary difference-in-differences: it measures how much a treated brand's outcome changed before and after its resale launch, then subtracts the same before-and-after change seen among control brands. Not every control brand counts equally, though, and neither does every pre-launch quarter. The estimator assigns weights to both, choosing them to match the treated brand's pre-launch trajectory as closely as possible. Formally, the treatment effect is estimated as:

$$\hat{\tau}_b = (\bar{Y}_b, post - \Sigma_t \lambda_t Y_b, t) - \Sigma_j \omega_j (\bar{Y}_j, post - \Sigma_t \lambda_t Y_j, t)$$

- $\bar{Y}_{bt}$, is the average star rating, sustainability score, or value VADER score for brand b in quarter t.
- b indexes the treated brand currently being estimated.
- j indexes the 27 control brands.
- t indexes quarter.

- ω_j is the weight assigned to control brand j, with $\omega_j \geq 0$ and $\Sigma_j\ \omega_j = 1$, chosen so the weighted average pre-launch path of the controls tracks brand b's own pre-launch path as closely as possible.
- λ_t is the weight assigned to pre-launch quarter t, with $\lambda_t \geq 0$ summing to one over the pre-launch period, chosen so the weighted pre-launch average is comparable to the post-launch average.
- $\bar{Y}_b, post$ is brand b's average outcome across its post-launch quarters; $\Sigma_t \lambda_t\ Y_b, t$ is its weighted pre-launch baseline.
- $\bar{Y}_j, post$ and $\Sigma_t \lambda_t\ Y_j, t$ are the equivalent terms for control brand j.
- $\hat{\tau}_b$ is the estimated treatment effect for brand b.

Each of the 19 treatment brands gets its own estimate, always compared against the same pool of control brands. Those individual estimates are then combined into a single overall number per outcome, giving more weight to the brands where the estimate is more precise. Standard errors come from the placebo method described in Arkhangelsky et al. (2021).

4.3.3 Study 2: Moderation analysis

Study 2 uses the brand as the unit of analysis. Rather than looking at individual reviews, it asks whether four design characteristics, integration level, messaging intensity, discount depth, and service offerings, explain why some brands responded to their resale launch more than others.

The design is cross-sectional and observational. Using raw post-launch ratings as the outcome was not workable because pre-launch means varied by more than two stars across brands, and that spread would contaminate any comparison. A simple before/after change score helps but not enough, since a brand's improvement would still be mixed up with was happening to ratings across the board during the same period. The outcome variable is therefore each brand's treatment effect from Study 1's synthetic difference-in-differences, which already accounts for both the brand's pre-launch baseline and the concurrent trend in

its synthetic control. What remains is the variation that plausibly reflects the resale launch itself. The estimation equation is:

$$\hat{\tau}_b = \beta_0 + \beta_1 Integration_b + \beta_2 Messaging_b + \beta_3 Discount_b + \beta_4 Service_b + \gamma_1 Outdoor_b + \gamma_2 Premium_b + \varepsilon_b$$

- $\hat{\tau}_b$ is brand b's treatment effect from Study 1, for whichever outcome (star rating, sustainability score, or value VADER score) is being modelled.
- $Integration_b$, $Messaging_b$, $Discount_b$, $Service_b$ are brand b's values on the four design characteristics.
- $Outdoor_b$ and $Premium_b$ are dummy variables for the outdoor-athletic and premium-fashion segments; contemporary fashion, the most common segment in the sample, is the omitted reference category.
- ε_b is the error term.

Each outcome is estimated separately using standard OLS regression, with 19 brands in the sample. Because some brands' treatment effects from Study 1 are estimated more precisely than others, depending on how many reviews they had, standard errors are corrected for this unevenness using the HC1 method (MacKinnon and White, 1985).

A second specification adds an interaction term between $Integration_b \times Messaging_b$, to test whether the effect of integration on the rating treatment effect depends on how strongly the brand frames its resale programme around sustainability. Construal level theory (Trope and Liberman, 2010) suggests that tightly integrated resale makes price comparisons harder for customers to ignore, while strong sustainability messaging may take some of that edge off by giving the resale programme a different meaning. A negative interaction coefficient would mean that having both high integration and strong messaging at the same time produces a smaller effect than you would expect from looking at each variable on its own.

5. Results

5.1 Descriptive Statistics

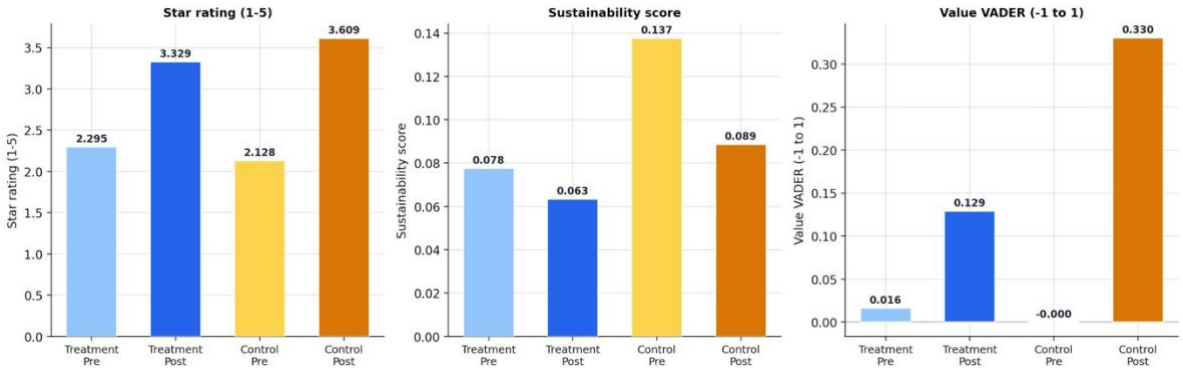
Table 2 breaks down the three outcome variables by group and period. The control group figures reflect the raw donor pool averages; the synthetic DiD estimator later reweights these brands to construct a closer counterfactual for each treatment brand. For control brands, the pre/post split uses the pseudo-event date each brand inherited from its assigned treatment brand within the same segment. Figure 1 shows the same comparison visually.

Table 2. Preliminary Evidence: Outcome Variables by Group and Period

Group / Period	n	Mean Rating	SD Rating	Mean Sust. Score	SD Sust. Score	Mean VADER	SD VADER
Treatment: pre-launch	5,457	2.294	1.772	0.078	0.308	0.016	0.563
Treatment: post-launch	21,399	3.329	1.868	0.063	0.294	0.129	0.510
Control: pre-event	3,798	2.128	1.706	0.137	0.456	0.000	0.593
Control: post-event	19,082	3.609	1.797	0.089	0.343	0.330	0.544

Note: Pseudo-event dates for control brands are inherited from matched treatment brands. Sust. Score = net sustainability dictionary score. VADER = compound sentiment score applied to value-related sentences; range [-1, 1]. Rating = Trustpilot star rating, 1-5 scale. SD = standard deviation. Means are computed across all individual reviews; brands with more reviews contribute proportionally more to the group average.

Figure 1. Pre vs Post Outcomes: Treatment vs Control



Treatment brands have a mean rating of 2.295 before the launch dates and control brands sit slightly lower at 2.128. Both groups improve afterward, but not by the same amount:

treatment brands rise from 2.295 to 3.329, a gain of 1.034, while control brands rise from 2.128 to 3.609, a gain of 1.481. The control group's improvement is, if anything, the larger of the two. Taken at face value that would suggest the resale launch made things worse, which makes little sense and points instead to the two groups not being comparable, rather than to a genuine negative effect. Part of the reason is who is in each group: the treatment group is dominated by premium and contemporary fashion labels, the kind that draw Trustpilot reviewers with grievances about counterfeits, late deliveries, and service failures, while the control group is made up of niche labels such as Organic Basics, Rab, Montane, and Le Chameau, whose customers tend to be happier reviewers to begin with. A raw pre/post comparison like this one cannot separate a true treatment effect from these pre-existing differences in who is being compared. Section 4.3.2 addresses this directly: the synthetic difference-in-differences estimator reweights the control group to match each treatment brand's own pre-launch trajectory, rather than relying on the raw, unweighted comparison shown here.

Value sentiment moves in the same direction. Treatment brands score 0.016 before launch; controls score 0.000[3] pre-event. Both increase post-event, with controls reaching 0.330 and treatment brands 0.129.

The sustainability score declines slightly after the event date for both groups. Control brands have a higher mean sustainability score pre-event (0.137) than treatment brands before their resale launch (0.078) or after (0.063), as shown in Figure 1. This is consistent with the same baseline imbalance already noted for star ratings and value sentiment: control brands like Organic Basics (37.2%) and Rab (28.0%) are built around environmental responsibility and attract customers who mention sustainability more frequently, while treatment brands draw customers who write about complaints (full breakdown by brand in Appendix A.5). Taken together with the rating and value-sentiment patterns above, this is further evidence that the

[3] The pre-event control mean of 0.000 reflects rounding to three decimal places; the unrounded value is −0.0001, close to zero because individual review-level value sentiment scores in this subsample are roughly symmetric between positive and negative, with a meaningful share (≈12%) scoring exactly neutral where no value-related language was detected."

raw control group differs from the treatment group on more than the resale launch alone, which is why Section 4.3.2's synthetic difference-in-differences estimator, rather than a simple comparison of these raw groups, is used for the main analysis.

LDA Topic Structure

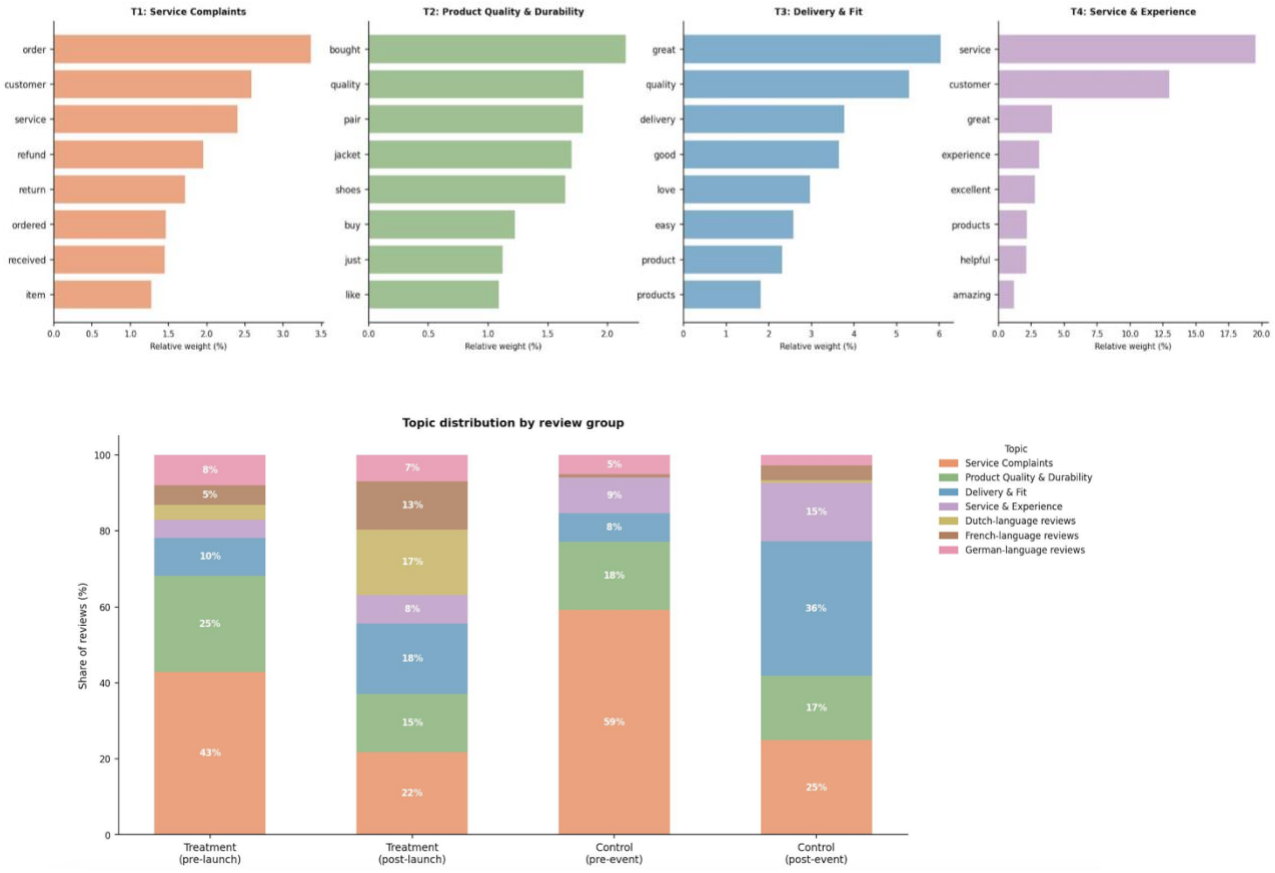
The seven-topic LDA model was run on 49,701 reviews as a diagnostic: to check what customers actually write about and whether the dictionary-based outcomes are picking up content that is genuinely there. The model returned four substantive English-language topics and three language-specific clusters, shown in the Figure 2.

The four English topics each cover a different part of the review experience. Delivery and Fit (10.0% of pre-launch treatment reviews, 18.5% post-launch; 7.5% control pre-event, 35.5% post-event) covers receiving and fitting purchased items. Service Complaints (42.8% pre-launch, 21.7% post-launch; 59.2% control pre-event, 24.9% post-event) groups negative post-purchase reviews around refunds, money disputes, and poor communication. Product Quality and Durability (25.4% pre-launch, 15.3% post-launch; 18.0% control pre-event, 16.9% post-event) covers material quality, construction, and how long products last. This is the topic most directly linked to the value perceptions the VADER outcome is designed to capture. Its share drops for treatment brands after the launch, control brands almost stay the same. Post-launch treatment reviews come less from customers writing about quality and more from customers writing about delivery or service interactions, though the topic data alone cannot establish why. It could reflect a change in who is writing reviews, or it could reflect the resale programme itself changing customers' actual experience, for instance through changes to delivery logistics or service load. Either way, this complicates how the VADER results should be read, since a change in sentiment after launch may partly reflect this shift in what customers are writing about, rather than a uniform change in how customers feel about the brand overall. Service and Experience (4.7% pre-launch, 7.5% post-launch; 9.3% control pre-event, 15.3% post-event) covers positive service interactions and general satisfaction with the purchase.

The three remaining clusters are language-based rather than thematic: Dutch-language reviews (largely Studio Anneloes customers), French-language reviews (Ba&sh, Maje,

Sandro, Balzac Paris), and German-language reviews (Rab and Hugo Boss predominantly). They show up because the brand set is multinational, not because they represent any meaningful content pattern.

Figure 2. LDA topics



The clearest pattern in the topic distribution is what happens to Service Complaints. Before any event date, both groups are dominated by dissatisfied customers: 43% of pre-launch treatment reviews and 59% of pre-event control reviews fall into this topic. Both then drop sharply, to 22% post-launch for treatment brands and 25% post-event for control brands, a decline of similar size in both groups. This fits the baseline imbalance already noted: before their respective event dates, both groups were dominated by dissatisfied customers writing about refunds and service failures.

After the launch, Delivery and Fit rises from 10.0% to 18.5% and Service and Experience from 4.7% to 7.5%, suggesting a broader range of customers writing about a wider set of

experiences. The post-event increase in Delivery and Fit for control brands (7.5% to 35.5%) reflects the same dynamic: Studio Anneloes accounts for the bulk of control post-event reviews and its customers write heavily about delivery.

It is important to note that the LDA findings are diagnostic, not causal because they describe what customers write about, not why outcomes change. The two groups wrote about different things before and after their event dates, and that difference matters for reading the dictionary-based measures: when the reviewer pool shifts, the sentiment and sustainability scores shift with it.

5.2 Study 1 Results

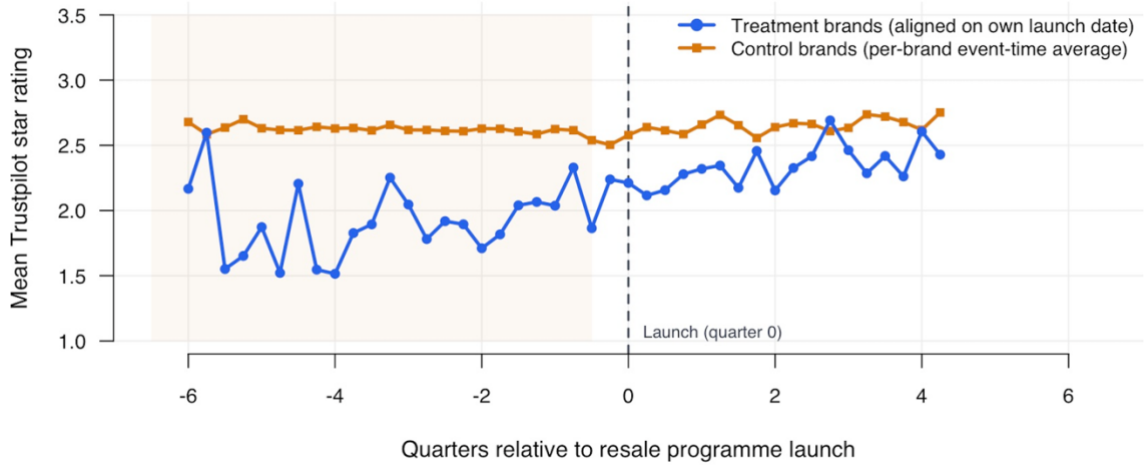
Baseline imbalance

Section 5.1 already shows that treatment and control brands differ in more than just whether they launched a resale programme: control brand customers write more positive reviews on average, mention sustainability more often, and discuss different topics before either group's event date. That composition difference, not a violation of the parallel trends assumption, is the main threat to a standard difference in differences design here.

Figure 3 shows the comparison directly. Averaging at the brand-quarter level and aligning each brand to its own launch date, treatment brands average 1.96 stars in the pre-launch window against 2.62 for control brands. These figures differ from the review-level means in Table 2 because each brand-quarter receives equal weight here regardless of review volume, consistent with how the synthetic DiD analysis aggregates the data. Both approaches confirm the same underlying story: the two groups start from different baselines. The two series move in a broadly similar pattern over time rather than steadily drifting apart, which points to a difference in starting level rather than a difference in trend. A simple pooled test, fitting one calendar time trend per group, suggests otherwise (interaction coefficient of -0.124, $p < 0.001$), but pooling brands that launched years apart onto a single calendar time axis can manufacture an apparent trend difference from staggered timing alone. Once pre-trends are tested within each launch cohort and aligned to each brand's own event time, as in the

Callaway and Sant'Anna (2021) check reported below, no significant pre-treatment divergence remains.

Figure 3. Mean Star Rating by Event Time: Treatment vs Control



Note: Blue shows treatment brands' own average Both lines show brand-quarter averages, giving each brand equal weight per quarter regardless of review volume. Treatment brands are each aligned to their own resale launch date. The control average at each event time is computed as the mean across all 27 control brands in the calendar quarter corresponding to that event time for each treatment brand, then averaged across all 19 treatment brands. Only event times with at least 6 brands contributing are shown.

This changes what the main concern actually is. The standard DiD design here is not threatened by diverging pre trends, it is threatened by the fact that the two groups are not drawing from the same kind of customer to begin with. Pre launch, control brand reviews are more heavily weighted toward Service Complaints (59%) than treatment brand reviews (43%), consistent with the two groups attracting different reviewer types from the outset rather than simply performing differently over time. That composition gap is what the synthetic difference in differences approach addresses directly, by reweighting the control group to match each treatment brand's own pre launch level and trajectory, rather than relying on the raw, unweighted comparison shown in Figure 3.

Two-Way Fixed Effects and Callaway-Sant'Anna (2021)

As a starting point, the two-way fixed effects DiD gives small, statistically indistinguishable from zero estimates for all three outcomes (star rating $\beta = 0.244$, $p = 0.301$; sustainability score $\beta = 0.005$, $p = 0.822$; value VADER $\beta = 0.022$, $p = 0.680$; full estimates in Appendix

A.6). The Callaway and Sant'Anna (2021) estimator, which accounts for staggered launch timing, gives the same null pattern but with a sign reversal for star rating (ATT = -0.337), consistent with the TWFE estimate being inflated by the comparability problems. Neither estimator accounts for the difference in customer composition between treatment and control brands, so neither result is treated as informative on its own.

These pooled averages also mask substantial variation across brands. Figure 4 shows the synthetic DiD treatment effect for star rating separately for each of the 19 treatment brands: effects range from -1.83 for Lululemon to +1.40 for Armedangels, with several individually significant in both directions. This heterogeneity, not the pooled estimate, is the more informative pattern, and it is what Study 2 investigates: whether differences in programme design explain why some brands' perceptions shift more than others.

Figure 4. Synthetic DiD Treatment Effect per Brand (Star Rating)

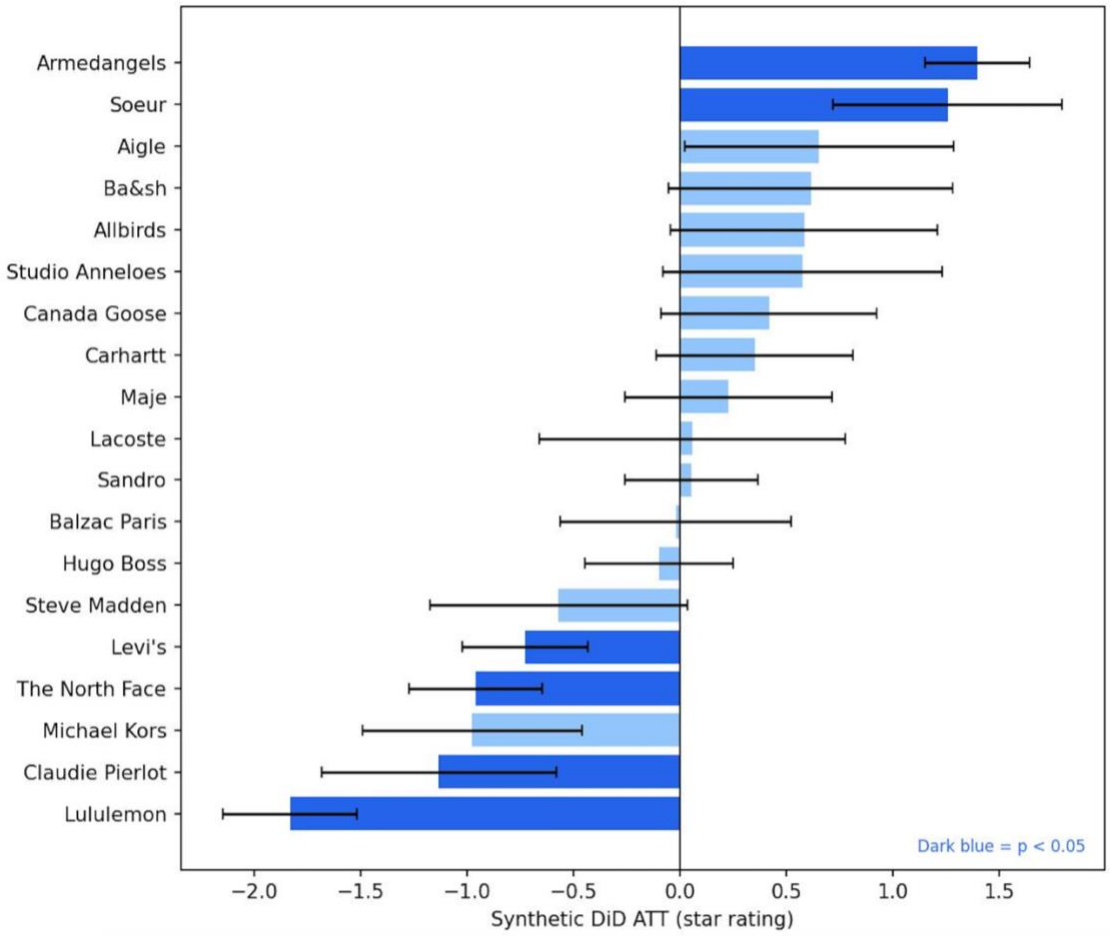
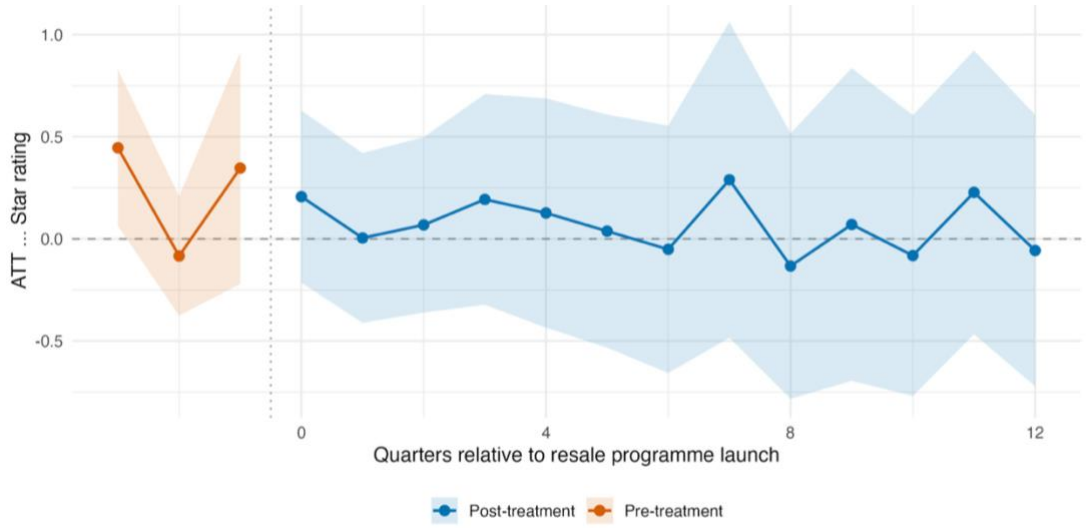


Figure 5 shows the Callaway and Sant'Anna event study referenced in the Baseline Imbalance discussion above. It confirms that the apparent pre-trend divergence found in the naive pooled test does not hold up once tested within launch cohorts.

Figure 5. Callaway & Sant'Anna (2021) Event Study: Star Rating



Note: Each point represents the cohort-weighted average treatment effect on star rating relative to one year before resale programme launch (reference period = quarter −4). Orange = pre-treatment placebo periods; blue = post-treatment periods. Shaded bands are 95% bootstrap confidence intervals. The absence of significant pre-treatment divergence supports the parallel trends assumption.

Resolving the trend question still leaves the comparability problem documented in Section 5.1 unaddressed, which is what the synthetic DiD approach below is designed to fix.

Synthetic DiD (Arkhangelsky et al., 2021)

Synthetic DiD addresses the comparability problem directly. For each of the 19 treatment brands, the estimator builds a custom weighted average of the 27 control brands whose pre-launch trajectory best matches that of the treatment brand, then applies a DiD adjustment on top. Table 3 reports the weighted average ATTs across all three outcomes alongside the TWFE and C&S estimates. Appendix A.7 tests whether these estimates are sensitive to the inclusion of four low-volume control brands.

Table 3. Study 1: Robustness Check Comparison Across Three Estimators

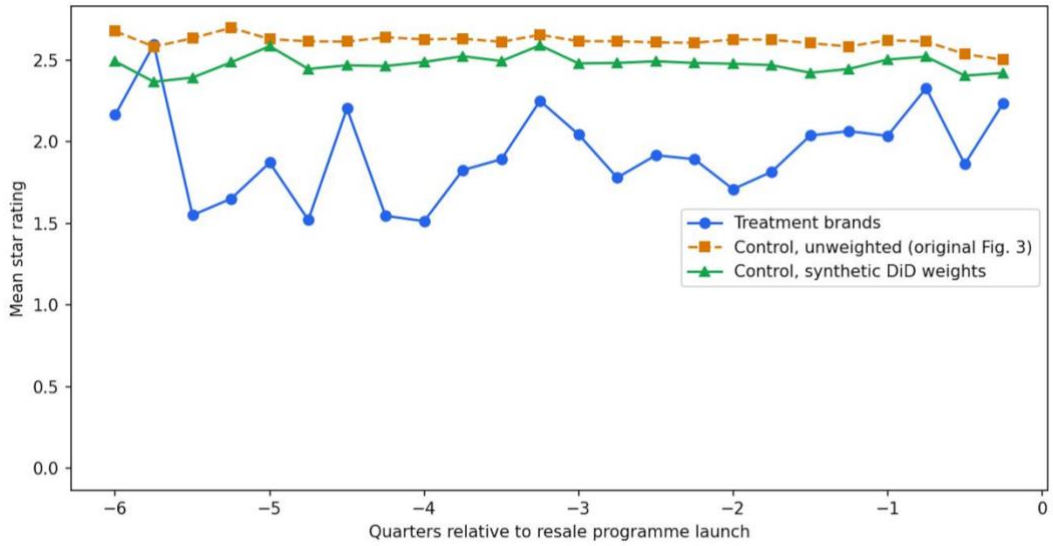
Outcome	TWFE β (p)	C&S ATT (p)	Synthetic DiD ATT (p)
Star rating	+0.244 (0.301)	−0.337 (0.225)	−0.078 (0.419)

Outcome	TWFE β (p)	C&S ATT (p)	Synthetic DiD ATT (p)
Sustainability score	+0.005 (0.822)	−0.014 (0.427)	−0.018 (0.052*)
Value VADER	+0.022 (0.680)	−0.052 (0.328)	−0.103 (0.002**)

*Note: Synthetic DiD estimated brand by brand with inverse-variance aggregation following Arkhangelsky et al. (2021); p-values from placebo method. *p < 0.10, **p < 0.05.*

Synthetic DiD's reweighting improves the pre-launch match only partially. Figure 6 compares the original unweighted control average from Figure 3 against the brand-specific synthetic control average, built using each treatment brand's own estimated weights. The mean absolute gap between treatment brands and the control group falls from 1.15 to 1.03 stars once the synthetic weights are applied, a reduction of about 10%, but a substantial level gap remains even after optimal reweighting: treatment brands sit consistently lower than the synthetic control throughout the pre-launch window. Although overall group means in Table 2 appear similar, the pre-launch gap reported here is computed brand by brand at each event-time quarter: individual brands vary substantially in their baseline rating levels, and the 27-brand donor pool cannot always reproduce those extremes through reweighting alone. The same partial improvement holds for the other two outcomes: for sustainability score, the pre-launch gap falls from 0.129 to 0.115 (an 11% reduction); for value VADER, from 0.304 to 0.286 (a 6% reduction). Across all three outcomes, a residual gap remains after reweighting, which means the donor pool incompletely reproduces the treatment brands' pre-launch trajectory regardless of the outcome considered. The aggregate gap also masks variation across brands: per-brand synthetic fits range from 0.64 stars (Studio Anneloes) to 1.58 stars (Allbirds), with a mean of 1.03 and standard deviation of 0.22. Brands with stronger pre-launch matches include some of the most credible estimates in Table 5: Levi's and Lululemon, two of the best-matched brands, show the largest and most significant negative ATTs. For the least well-matched brands, particularly Allbirds, Sandro, and Balzac Paris, the ATT estimates should be interpreted with more caution.

Figure 6. Pre-launch fit: Raw Control vs Synthetic-weighted Control



Note: Blue shows treatment brands' own average pre-launch rating. Orange shows the original unweighted average across all 27 control brands, aligned per treatment brand to their own event time, as in Figure 3. Green shows the synthetic DiD control average, built from each treatment brand's own estimated weights and then averaged across all 19 treatment brands. The mean gap between treatment brands and the control group falls from 1.15 to 1.03 stars under synthetic weighting, a reduction of about 10%. The residual gap is a level difference rather than a widening trend, the gap stays roughly stable across the pre-launch window rather than growing, which is the relevant condition for the DiD estimates below to be valid.

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All three synthetic DiD estimates point in the same direction: negative. The rating result is negative and non-significant (ATT = -0.078, p = 0.419). The sustainability score is marginally negative (ATT = -0.018, p = 0.052), just above the conventional significance threshold. The value VADER result is negative and significant (ATT = -0.103, p = 0.002). None of these are positive, and that consistency across three independently estimated outcomes is itself informative, even where individual estimates fall short of significance: the best available evidence points toward resale launches being associated with a decline in customer perception, not an improvement. This contrasts with TWFE, which gave a positive estimate for value VADER, suggesting that estimate was inflated by the pre-treatment imbalance documented above. Once that imbalance is corrected, the value sentiment decline becomes detectable, and it is significant.

Segment-level results

Table 4 breaks down the synthetic DiD ATT estimates by segment. The pattern is more informative than the pooled estimates: outdoor and athletic brands show a significant negative effect across all three outcomes (rating ATT = -0.780, $p < 0.001$; sustainability ATT

= -0.066, p < 0.001; value VADER ATT = -0.294, p < 0.001), while contemporary fashion brands show a significant positive rating effect (ATT = +0.297, p = 0.022) alongside a significant negative sustainability effect (ATT = -0.037, p = 0.009). Premium fashion shows a significant positive sustainability effect (ATT = +0.080, p < 0.001) but no significant effect on rating or value VADER. These results still call for caution: there are only 10 treatment brands in contemporary fashion, 5 in outdoor and athletic, and 4 in premium fashion, small numbers for any single segment. Segment is a structural control here, not a theoretically motivated moderator; how the effect varies with integration level, messaging intensity, discount depth, and service offerings is examined directly in Study 2.

Table 4. Study 1: Segment-Level Synthetic DiD ATT Estimates

Segment	Outcome	ATT	SE	t	p	N (brands)
Contemporary Fashion	Star rating	0.297	0.129	2.300	0.022	10
Contemporary Fashion	Sustainability score	-0.037	0.014	-2.603	0.009	10
Contemporary Fashion	Value VADER	-0.017	0.046	-0.373	0.709	10
Outdoor & Athletic	Star rating	-0.780	0.182	-4.277	<0.001	5
Outdoor & Athletic	Sustainability score	-0.066	0.017	-3.905	<0.001	5
Outdoor & Athletic	Value VADER	-0.294	0.063	-4.698	<0.001	5
Premium Fashion	Star rating	-0.155	0.237	-0.656	0.512	4
Premium Fashion	Sustainability score	0.080	0.019	4.098	<0.001	4
Premium Fashion	Value VADER	-0.066	0.071	-0.927	0.355	4

Note: Synthetic DiD ATTs aggregated by segment using inverse-variance weights.

Brand-level results

Table 5 presents the raw pre-to-post mean changes in all three outcomes for each treatment brand, ordered by rating change, with each brand's synthetic DiD ATT for direct comparison. The Δ columns are unconditional, unadjusted comparisons: just that brand's own post-launch average minus its own pre-launch average, with no control group or correction of any kind. They describe what happened to each brand's Trustpilot trajectory, not why.

Table 5. Brand-Level Pre-to-Post Changes in Outcome Variables

Brand	Δ Rating	Δ Sustainability	Δ Value VADER	Synth. DiD ATT (Rating)	Synth. DiD ATT (Sust.)	Synth. DiD ATT (VADER)
Canada Goose	2.164	-0.117	0.432	0.417	0.135	0.288
Ba&sh	1.912	-0.047	0.096	0.614	0.105	0.182
Balzac Paris *	1.812	0.011	0.534	-0.020	-0.018	0.063
Aigle	1.112	-0.021	0.050	0.652	0.025	-0.047
Soeur	0.756	-0.023	0.064	1.257	-0.040	0.022
Maje	0.716	0.013	0.178	0.226	-0.123	-0.101
Steve Madden	0.539	0.000	0.070	-0.569	-0.070	0.093
Sandro	0.237	-0.009	0.084	0.052	-0.117	0.314
Lacoste	0.191	0.063	0.072	0.059	0.158	0.017
Studio Anneloes	0.133	0.001	0.021	0.575	0.032	0.033
Claudie Pierlot *	0.089	-0.021	0.131	-1.132	-0.005	0.265
Carhartt	-0.092	0.049	0.015	0.351	-0.225	-0.223
Michael Kors	-0.098	0.042	-0.010	-0.976	0.063	-0.038
Levi's	-0.150	0.034	-0.044	-0.728	0.022	-0.339
Armedangels	-0.153	-0.006	-0.152	1.398	-0.029	0.350
Lululemon	-0.305	-0.009	-0.060	-1.831	0.011	-0.328
Hugo Boss	-0.342	-0.001	-0.178	-0.099	0.019	-0.288
Allbirds	-0.386	-0.041	-0.152	0.583	-0.153	-0.092
The North Face *	-0.886	-0.217	-0.338	-0.960	-0.059	-0.544

*Note: * = borderline DiD eligibility (one window below 50-review threshold). Δ = post-launch mean minus pre-launch mean (unconditional, not causally adjusted). Synth. DiD ATT = the same brand's synthetic difference-in-differences estimate underlying the pooled results in Table 3 and the rating chart in Figure 4. Large gaps between a brand's raw Δ and its ATT (e.g., Balzac Paris, Armedangels, Soeur) indicate the raw change is substantially driven by factors other than the resale launch, such as regression to the mean or a divergent synthetic control trajectory.*

The brand-level results vary considerably, and the gap between raw and adjusted numbers is often large. Canada Goose, Balzac Paris, and Ba&sh all improved by more than 1.5 stars in raw terms. Those three brands also had the lowest pre-launch baselines, which raises the question of how much of that improvement is regression to the mean rather than any resale effect.

The adjusted numbers put this in perspective. Canada Goose and Ba&sh still show positive effects after correction (+0.417 and +0.614), smaller than the raw changes but still there. Balzac Paris tells a different story: its synthetic DiD ATT is essentially zero (-0.020), meaning almost none of its 1.9-star raw improvement survives once the counterfactual is accounted for. The pattern also runs the other way. Armedangels shows a small raw decline (-0.153) but the largest positive adjusted effect in the table (+1.398). Soeur's modest raw gain (+0.756) grows into an even larger adjusted effect (+1.257). The North Face's large negative raw change (-0.886) is partly a product of its June 2018 launch date, which places most of its review activity during the COVID period when outdoor brand service was widely disrupted. Its adjusted ATT (-0.960) confirms the decline holds even after correction. The raw column shows what happened to each brand. The adjusted column shows how much of that can possibly be traced back to the resale launch.

The same table shows that for sustainability score, 10 of the 19 brands show a decline after launch. The LDA topic analysis provides context: even post-launch, the Product Quality and Durability topic accounts for only 15.3% of treatment reviews, limiting the baseline for detecting language shifts. For value VADER, ten brands show a positive post-launch change and nine a negative one.

5.3 Study 2 Results

Table 6 reports the synthetic DiD ATT for each brand regressed on the four design characteristics and segment, the specification described in Section 4.3.3. None of the four design characteristics reach conventional significance for any outcome. The star rating model's adjusted R^2 is negative (-0.275), indicating that with only 19 brands and 7

parameters, the model is overfit rather than informative. Testing each moderator individually, controlling only for segment, produces the same null pattern, so the result is not an artifact of collinearity among the four moderators in the joint model.

Table 6. Study 2: Synthetic DiD ATT Regressed on Design Characteristics

Variable	Synth. DiD ATT: Star Rating			Synth. DiD ATT: Sust. Score			Synth. DiD ATT: Value VADER		
	β	SE	p	β	SE	p	β	SE	p
Integration level	-0.210	0.478	0.661	0.014	0.055	0.803	-0.115	0.105	0.272
Messaging intensity	0.177	0.248	0.476	0.015	0.027	0.588	0.003	0.083	0.974
Discount depth (%)	-0.021	0.027	0.432	-0.001	0.003	0.649	-0.004	0.010	0.711
Service offerings	0.072	0.351	0.838	-0.041	0.048	0.393	-0.029	0.118	0.809
Outdoor & athletic (seg.)	-0.990	0.824	0.230	-0.061	0.066	0.356	-0.323	0.194	0.121
Premium fashion (seg.)	-0.805	0.546	0.141	0.101	0.057	0.074*	-0.139	0.157	0.392
Intercept	1.298	1.691	0.443	0.041	0.200	0.838	0.588	0.603	0.349
Adjusted R^2	-0.275			0.169			0.107		
SE type	HC1 robust			HC1 robust			HC1 robust		
N (brands)	19			19			19		

*Note: outcome is each brand's synthetic DiD ATT from Table 3/Figure 4, not a review-level DiD coefficient. *p < 0.10. No coefficient reaches p < 0.05.*

The null result does not mean design characteristics are unrelated to the treatment effect. It means 19 brands is not enough to detect that relationship reliably once the outcome is properly adjusted. Each design characteristic is fixed at the brand level, so the same value applies to every review a brand receives. The effective sample size for this test is therefore 19, no matter how many individual reviews sit underneath. Some imprecision at that scale is to be expected.

Interaction model

Construal level theory predicts that tightly integrated resale makes price comparisons harder to avoid, while strong sustainability messaging could soften that by changing what the resale launch means to the customer. Table 7 tests this by entering integration, messaging, and their interaction together in one model. One result holds up: with sustainability score as the outcome, the interaction term is significant. The same interaction on star rating does not reach significance and the adjusted R^2 is negative, so that model is not informative.

Table 7. Study 2. Integration × Messaging Interaction, by Outcome

Variable	Star Rating			Sust. Score			Value VADER		
	β	SE	p	β	SE	p	β	SE	p
Integration	1.354	0.897	0.152	0.211	0.070	0.009**	0.210	0.230	0.374
Messaging	1.200	0.712	0.113	0.163	0.072	0.039*	0.205	0.226	0.379
Integration × Messaging	-0.486	0.301	0.126	-0.070	0.026	0.016*	-0.113	0.085	0.205
Intercept	-3.315	2.152	0.144	-0.496	0.191	0.020*	-0.310	0.617	0.623
Adjusted R^2	-0.032			0.032			0.167		
N (brands)	19			19			19		

*Note: outcome is each brand's synthetic DiD ATT. HC1 robust standard errors in parentheses. *p < 0.05, **p < 0.01.*

Integration and messaging appear to work as substitutes, not complements. A brand that deeply embeds resale into its core shopping experience earns a sustainability-perception benefit without needing to message heavily around it. The structure does the talking. A brand that keeps resale more separate can still earn something similar through messaging instead. But combining both does not stack the two effects. It actually produces less than either alone would suggest. This points to a signalling backfire. Once the structural choice already signals genuine commitment, adding heavy sustainability language on top risks reading as overstatement. The message undermines itself.

With only 19 brands and multiple models tested across three outcomes, this result should be read as suggestive rather than confirmatory. It is reported here because it is the only

moderation pattern that holds up once the outcome is properly adjusted, not because it was predicted in advance. Appendix A.8 runs the same interaction on raw pre-to-post change scores instead. There, the interaction is significant for star rating rather than sustainability score. The two versions of the outcome produce different winners. That alone is a reason to be cautious about either result.

6. Discussion

6.1 Addressing 1st Research Question

The first question asked whether branded resale launches shift how customers perceive a brand, and in which direction. The honest answer is that the shift, where it exists, runs negative rather than positive. All three synthetic DiD estimates point the same way: rating falls (ATT = -0.078, not significant), sustainability language falls slightly (ATT = -0.018, just above the conventional threshold), and value sentiment falls and does so significantly (ATT = -0.103, $p = 0.002$). None of the three outcomes moved upward. That consistency matters more than any single p-value, because it is the kind of pattern that is hard to produce by chance across three independently estimated measures.

This is not the story the industry usually tells about branded resale. Patagonia's Worn Wear and similar programmes are routinely framed as proof that secondhand and new can coexist comfortably, even reinforce each other. The data here suggests something closer to the cannibalization concern that Bulow (1982, 1986) and Waldman (1997) described decades ago in a different market context, where the mere existence of a resale channel, not any particular way of presenting it, already changes how customers value the new product.

The value VADER result gives the clear picture. Customers write more negatively about price and value after a brand launches resale, but this only shows up once the comparison group is properly weighted to match each treatment brand's pre-launch trajectory. Under the unweighted comparison the effect disappears, and for TWFE it even flips sign. That gap

between the corrected and uncorrected estimate is itself worth noting: the simple before-and-after comparison was hiding a real decline by using the wrong baseline.

The sustainability result is weaker, and that is worth sitting with. The brand identity literature suggests customers should at least notice and reward an environmental gesture, even an imperfect one, because building resale infrastructure is a real commitment rather than a marketing claim (Spence, 1973). The sustainability score barely moves. When it does move, it goes the wrong way. That suggests the identity signal is not strong enough to override what is happening to price perception. Why? It could be that the signal is just too weak to register. It could also be that customers care more about price than environmental virtue in the specific moment they sit down to leave a Trustpilot review. Probably some of both, and this study cannot fully differentiate them. What the topic modelling does add is this: post-launch reviews shift away from product quality and toward delivery and service topics. Sentiment is not just changing within the same conversations. The conversations themselves are changing.

Underneath the pooled numbers, the brands are telling very different stories. Outdoor and athletic brands show a significant negative effect on all three outcomes simultaneously, the most consistently unfavorable pattern in the data. Contemporary fashion brands go the other way on rating, a significant positive effect, but still show a significant negative on sustainability language. Premium fashion lands closer to neutral on rating and value, with one exception: a significant positive sustainability effect. A single pooled number was never going to capture this much variation. It didn't. The segment breakdown is arguably more informative than the headline ATT. Whether resale helps or hurts customer perception depends heavily on what kind of brand is doing it, not just on the decision to launch.

6.2 Addressing 2nd Research Question

The second question asked which design choices determine whether resale works in a brand's favor. The answer is simpler. None of the four characteristics tested, integration level, messaging intensity, discount depth, or service offerings, predict the size of the effect. The

joint model returns no significant coefficient for any outcome. The star rating version has a negative adjusted R^2, what you get when seven parameters are fitted to nineteen brands.

Testing each characteristic separately produces the same null pattern, so collinearity is not the explanation. The four variables are not cancelling each other out. The issue is simpler: this question only has nineteen data points to answer it with. That is not enough to detect anything but a fairly large effect, and these effects are not large enough.

That null result should not be explained away. It does not mean integration, messaging, discount depth, and service offerings are irrelevant to how resale gets perceived. It means this sample cannot distinguish their individual contributions with any confidence. That is a different claim. A smaller one, but not a wrong one.

One pattern does survive, and it survives because no single variable carries the whole story. Entering integration, messaging, and their interaction together with sustainability score as the outcome produces a significant result on all three terms. Both moderations on their own are positive, but the interaction is negative, large enough to cancel out a meaningful share of either main effect when both are high simultaneously. A brand that builds deep structural integration into its resale programme earns a sustainability perception benefit without needing to say much about it. A brand that keeps resale more separate can reach a similar place through heavy sustainability messaging instead. Doing both at once does not add the two benefits together, instead they behave as substitutes. This fits a signalling backfire reading better than the original prediction. The conceptual framework drew on construal level theory and expected the interaction to show up in star rating, where price perception is most direct. It did not. The effect is in sustainability language. Once the structural choice has already communicated genuine commitment, heavy messaging on top just looks like overselling. Lyon and Maxwell (2011) describe exactly this: firms that talk about their own virtue more than necessary tend to generate skepticism rather than credit.

It is worth being direct about how much weight this single result can bear. One significant interaction, across nineteen brands and several models, is not confirmed theory. It survived one specific test. It did not survive the same test on the outcome it was originally predicted for. When the sensitivity check uses raw uncorrected change scores instead, the interaction

flips to star rating. The opposite pairing entirely. That divergence does not erase the finding, but it should keep it in proportion. This is a pattern worth following up, not a settled answer to the second research question.

7. Limitations

Most of the limitations here come from the same place: a small, imperfectly matched sample of brands observed through a single review platform.

The most consequential limitation is the control group. Even after synthetic DiD reweighting, the pre-launch gap between treatment brands and their synthetic controls only closes by about 10%. None of the 27 control brands starts from a baseline rating as low as the typical treatment brand. No combination of weights can fix that entirely. The gap stays stable across the pre-launch window rather than widening, which is the condition that matters most for a difference-in-differences design. But stable is not the same as closed. The direction of the headline findings is more trustworthy than the precise size of any individual estimate.

The same problem shows up in the review content itself. Before their respective launch dates, both treatment and control brands were dominated by dissatisfied customers writing about refunds and service failures. When both groups start from the same kind of dysfunction, the parallel trends assumption is hard to satisfy on its own, and this is separate from the rating-level gap already discussed. It is one more reason the synthetic difference-in-differences estimator is doing the heavy lifting here rather than a standard two-way fixed effects design.

A related issue sits in the rating scale itself. A brand near the bottom of a five-point scale has more room to improve than one starting in the middle. That is just how bounded scales work, not evidence of anything the resale programme did. With the control brands available here, this study cannot fully rule that out.

Study 2's sample size is its own constraint, separate from the comparability issue in Study 1. Nineteen brands is enough to estimate one effect at a time, which is why the segment-level and brand-level breakdowns hold up. Asking four design questions simultaneously and expecting a clean answer for each is a different matter. The negative adjusted R^2 in the joint

rating model is the most visible sign of that. Future replication with a larger sample would help. More retailers are launching first-party resale every year, and once enough of them have accumulated a usable review history, the same question can be asked with the statistical power it actually needs. Also, Study 2 has a deeper causal problem underneath the sample size issue. Integration level, messaging choices, these were set by each brand itself, not assigned at random. They almost certainly correlate with things this dataset cannot see: how sophisticated the brand's marketing operation is, where it sits in its category, how confident it feels about resale as a strategic bet. Using a causally adjusted outcome from Study 1 fixes the counterfactual problem. It does not fix the fact that the moderators are observational. A brand that deeply integrates resale into its core shopping experience may just be a fundamentally different kind of company than one that keeps it separate. This dataset cannot fully untangle that from the design choice itself.

The sensitivity check in Appendix A.8 makes this concrete. Swapping the causally adjusted outcome for each brand's raw unadjusted change score changes which interaction comes out significant. Sustainability under one specification, star rating under the other. Both cannot be right. This study uses the corrected version, and the reasoning behind that choice is laid out in the methods. But the fact that two reasonable specifications point in opposite directions is a limitation.

Lastly, the topic modelling results raise one more issue. Post-launch reviews lean more on delivery and service language and less on product quality than pre-launch reviews do, and this holds for both treatment and control brands. Some of the VADER decline documented in Study 1 may reflect that compositional shift rather than a genuine change in how customers feel about the brand. The sustainability measure has its own version of this problem. It has not been validated against human-coded data. A small number of false positives or false negatives in individual word matches could move a brand's score without any real change in how customers talk about sustainability underneath it.

Each limitation has a fairly specific fix, not just a request for more data. The comparability gap would narrow with a bigger donor pool, drawing on other review platforms or brands outside the segments used here to find control brands whose baseline ratings sit closer to the

treatment group. The power problem in Study 2 is partly just a matter of time: more first-party resale programmes are launching every year, and pooling across platforms in the meantime could help sooner. Causal ambiguity around the moderators is the harder problem, and observational data can only go so far with it. A field experiment manipulating integration level or messaging intensity for the same brand, even a single brand running a controlled test across markets, would let someone claim causal moderation rather than infer it from cross-brand differences. The sensitivity check disagreement and the topic composition issue both point toward the same underlying need: more brands, better measurement, and sentiment models that condition on what reviewers are actually writing about rather than treating all reviews as interchangeable.

8. Practical Implications

The first implication is the least comfortable one. Do not assume launching resale will improve how customers see you. Across nineteen brands the average effect leans negative, and the clearest finding in the data is a decline in how customers write about value and price. A brand expecting a sustainability benefit should prepare for the opposite. Putting a cheaper used version of the product next to the new one invites exactly the price comparison that makes the original feel less worth paying for. That risk lands harder on some brands than others. Outdoor and athletic brands show a significant negative effect across all three outcomes at once, the most consistently unfavorable pattern in the sample. A brand in that category should not assume the more mixed experience of premium or contemporary fashion brands will transfer. A limited pilot, one market or one product line, before a full rollout seems like the more sensible starting point.

The second implication is simpler. Integration and messaging do not add up together. The data does not support pushing both to maximum strength at once. A brand with deeply integrated resale, new and used items visible side by side, earns a sustainability perception benefit from that structural choice alone, no heavy messaging needed. A brand that keeps resale more separate can reach the same place through messaging instead. Doing both at once produced a smaller benefit than either approach alone. Customers seem to read the

combination as overselling once the structure has already made the point. Pick one and commit.

The third implication is about what this study cannot tell a manager. Discount depth and service offerings showed no detectable relationship with the size of the perception effect. That null result does not mean these choices are unimportant. It means 19 brands was not enough to find out. A manager reading this should not take the silence on discount strategy and service infrastructure as permission to deprioritise them. This study just does not have the data to weigh in either way.

9. Conclusion

This thesis asked a question the existing literature had largely left open: what happens to customer perception once a brand starts selling its own product secondhand. The answer, built from nearly 50,000 unsolicited Trustpilot reviews across 48 brands, is more cautionary than the industry tends to admit. Resale launches are associated with a decline in customer perception on average, not an improvement. The clearest signal is not in sustainability language but in how customers write about price and value. That effect is not uniform either. Outdoor and athletic brands fare worst, showing significant negative effects across all three outcomes at once, though this thesis cannot establish why beyond noting that these brands are typically positioned around durability and performance, a positioning a secondhand channel might call into question in ways this data cannot directly test. Contemporary fashion brands show a more mixed picture, with some improving on rating and others declining, suggesting the perception effect depends heavily on how the brand is positioned and what customers expect from it going in.

The second part of the study asked whether specific design choices explain that variation. The honest answer is mostly no, at least not individually, and not with nineteen brands. The one pattern that held up, that structural integration and sustainability messaging work as substitutes rather than complements in shaping sustainability perception, is a real finding and

theoretically coherent. But it is narrower than the four-lever framework originally proposed, and it should be treated as a starting point for further testing rather than a settled result.

This thesis also makes a methodological point. The standard difference-in-differences approach looked reasonable at first glance but rested on a control group that was not actually comparable to the treatment brands. Correcting for that through synthetic difference-in-differences did not refine the result. It reversed it. A positive, non-significant TWFE estimate for value sentiment became negative and significant once the comparison group was properly weighted. That kind of reversal does not happen often, but when it does it is a reminder that method choice is not a technical detail. It determines what a study can actually say.

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A. Appendix

A.1 Control Brand Overview

Table A.1 lists the 27 control brands, their segment, the treatment brands they are matched to, and their total review count. Control brands have no confirmed first-party resale programme. Four brands (Tommy Hilfiger, Fred Perry, Jottnar, and Vuori) have fewer than 150 reviews and are flagged as low volume; they are retained in the main specification and excluded in a robustness check.

Table A.1. Control Brand Overview

Brand	Segment	Matched to	Total Reviews	Volume flag
Zadig & Voltaire	CF	Ba&sh / Sandro / Maje	456	Marginal
Sézane	CF	Ba&sh / Sandro / Maje	1,855	OK
The Kooples	CF	Ba&sh / Sandro / Maje	720	OK
Wrangler	CF	Levi's	211	Marginal
Guess	CF	Steve Madden	878	OK
Organic Basics	CF	Armedangels	1,496	OK
Rouje	CF	Balzac Paris	1,084	OK
Tommy Hilfiger	PF	Hugo Boss / Lacoste	140	Low
Ted Baker	PF	Hugo Boss / Lacoste	1,780	OK
Armani	PF	Hugo Boss	264	Marginal
Fred Perry	PF	Lacoste	146	Low
Moncler	PF	Canada Goose	366	Marginal
Moose Knuckles	PF	Canada Goose	194	Marginal
Furla	PF	Michael Kors	198	Marginal
Tory Burch	PF	Michael Kors	261	Marginal
Columbia Sportswear	OA	Patagonia	748	OK
Rab	OA	Patagonia / Arc'teryx	1,172	OK

Brand	Segment	Matched to	Total Reviews	Volume flag
Berghaus	OA	Arc'teryx / The North Face	1,746	OK
Salomon	OA	Arc'teryx / Carhartt	1,127	OK
Montane	OA	Arc'teryx	1,778	OK
Jottnar	OA	Arc'teryx	99	Low
Vuori	OA	Lululemon	99	Low
Under Armour	OA	Lululemon	959	OK
Hoka	OA	Allbirds	1,832	OK
Dickies	OA	Carhartt	172	Marginal
Le Chameau	OA	Aigle	1,817	OK
Osprey	OA	The North Face	1,282	OK

Note: CF = Contemporary Fashion; PF = Premium Fashion; OA = Outdoor & Athletic. Low = fewer than 150 total reviews (Jottnar, Vuori, Tommy Hilfiger, Fred Perry); Marginal = 150–499 reviews.

A.2 Sustainability Dictionary

The sustainability language score is computed by applying two dictionaries to the combined title and body of each review. The net score equals positive matches minus negative matches. All matching is stem-based (* denotes a wildcard suffix).

Table A.2a. Positive Dictionary (38 stems and patterns)

Category	Terms
Environmental practice	sustain*, recycl*, circular, carbon, environmental, responsible, ethical, conscious, organic, biodegradable, zero-waste, low-impact, fair-trade
Circular product flows	resale, second-hand, secondhand, pre-owned, pre-loved, upcycl*, repair, reuse, renew*
Product longevity	durable*, durability, longevity, lifecycle
General environmental framing	green, eco, climate, planet, waste

Table A.2b. Negative Dictionary (7 stems targeting greenwashing scepticism)

Stem	Target expression
greenwash*	Explicit greenwashing accusation
hypocri*	Hypocrisy claims regarding environmental statements
performativ*	Performative or virtue-signalling framing
virtue-signal*	Virtue signalling
mislead*	Misleading environmental claims
deceptive	Deceptive environmental framing
fake-green	Explicit fake sustainability label

Note: The dictionary was constructed deductively from the circular economy literature and the sustainability communication framework of Delmas and Burbano (2011). It has not been validated against a human-annotated ground truth; false positives are possible (e.g., "natural" describing fabric composition, "repair" describing a product defect). Null sustainability scores should not be interpreted as evidence that sustainability perceptions are absent.

A.3 Treatment Brand Moderator Coding

Table A.3 presents the four Study 2 moderator variables coded for each of the 21 treatment brands, along with the resale platform, launch date, and review window counts. All four variables were coded from primary sources (brand resale platforms, press releases, and marketing materials) prior to and independently of the outcome data. Integration level and messaging intensity are five-point scales, service offerings is a count variable (maximum 3), and discount depth is the mean percentage discount across five matched resale items.

Table A.3. Treatment Brand Moderator Scores and Sample Counts

Brand	Segment	Platform	Launch	Integration	Messaging	Discount(%)	Discount SD	Services	Pre n	Post n	DiD Status
Ba&sh	CF	Faume	Nov 2022	3	1	53.4	9.4	1	65	568	Included
Sandro	CF	Faume	Oct 2021	2	1	63.6	7.2	1	116	235	Included
Maje	CF	Faume	Oct 2022	2	1	63.7	7.3	1	106	188	Included
Claudie Pierlot	CF	Faume	Feb 2024	2	2	59.5	4.8	1	70	40	Borderline
Balzac Paris	CF	Faume	Nov 2021	2	2	47.3	22.8	1	32	1,448	Borderline
Soeur	CF	Faume	Feb 2025	3	3	52.4	11.6	2	284	298	Included
Steve Madden	CF	Trove	Jan 2022	3	3	51.9	17.6	2	92	1,145	Included
	CF	Trove	Mar 2022	2	5	66.9	19.7	2	479	1,011	Included
Studio Anneloes	CF	Archive	Feb 2024	2	3	48.8	7.3	1	197	3,208	Included

Brand	Segment	Platform	Launch	Integration	Messaging	Discount(%)	Discount SD	Services	Pre n	Post n	DiD Status
Levi's	CF	Trove	Oct 2020	2	4	57.1	11.7	2	288	586	Included
Hugo Boss	PF	Faume	Apr 2022	3	4	44.5	7.3	2	366	508	Included
Lacoste	PF	Faume	Dec 2024	2	3	48.6	5.7	1	557	91	Included
Canada Goose	PF	Trove	Jan 2023	2	4	33.6	7	3	434	2,047	Included
Michael Kors	PF	Trove	Aug 2022	2	3	47.0	8	1	679	412	Included
Lululemon	OA	Trove	Apr 2022	3	4	45.5	13.3	2	611	595	Included
Allbirds	OA	Trove	Feb 2022	3	5	41.8	6.2	2	549	1,451	Included
Aigle	OA	Faume	Oct 2020	2	4	45.6	11	2	66	953	Included
Carhartt	OA	Trove	Mar 2023	2	4	34.9	10.6	3	425	241	Included
The North Face	OA	Archive	Jun 2018	3	4	50.1	10.1	3	39	2,008	Borderline
Arc'teryx	OA	Trove	May 2019	3	4	31.3	5.3	3	0	4,000	Excluded
Patagonia	OA	Trove	2013	5	5	42.7	7.1	3	2	366	Excluded

Note: CF = Contemporary Fashion; PF = Premium Fashion; OA = Outdoor & Athletic. Integration = integration level (1–5); Messaging = messaging intensity (1–5); Discount = mean discount depth across 10 matched items; Discount SD = standard deviation of discount depth across the ten matched listings, reflecting within-brand price dispersion. Services = service offerings count (1–3). DiD status: Included = pre ≥ 50 and post ≥ 50; Borderline = one window below threshold. Michael Kors and Steve Madden operate peer-to-peer platforms; discount depth reflects market pricing.

A.4 LDA Topic Model: Technical Details

The seven-topic solution was chosen by running models with five to ten topics and comparing held-out perplexity and manual coherence. Lower perplexity means better statistical fit; coherence measures whether the top words in each topic hang together in a way that makes sense. Table A.4a reports perplexity across all solutions. Seven topics gave the biggest coherence improvement over six while staying fully interpretable. Eight or more topics started splitting into fragments without any clear meaning.

Table A.4a. LDA Model Selection: Perplexity by Number of Topics

Number of topics	Held-out perplexity	Interpretable?
5	2,841	Partially, topics overlap substantially
6	2,703	Partially, language clusters not separated
7	2,618	Yes, selected solution
8	2,589	Partially, two English topics fragment
9	2,571	No, topic fragmentation
10	2,558	No, topic fragmentation

Table A.4b. Seven-Topic Solution: Top 10 Terms per Topic

Topic	Label	Top 10 terms (by relative weight)
1	Service Complaints	order, customer, service, refund, return, ordered, received, item, days, online
2	Product Quality & Durability	bought, quality, pair, jacket, shoes, buy, just, like, store, purchased
3	Delivery & Fit	great, quality, delivery, good, love, easy, product, products, fast, quick
4	Service & Experience	service, customer, great, experience, excellent, products, helpful, amazing, support, team
5	Dutch-language reviews	en, ik, snelle, levering, een, het,
6	French-language reviews	et, le, la, je, est, dã, en, que, trã, une

Topic	Label	Top 10 terms (by relative weight)
7	German-language reviews	und, ich, die, der, nicht, ist, sehr, habe, das, den

Note: Model estimated using scikit-learn online variational Bayes (Pedregosa et al., 2011), 30 iterations, learning offset = 50, random seed = 42.

Table A.4c. Topic Share by Review Group (%)

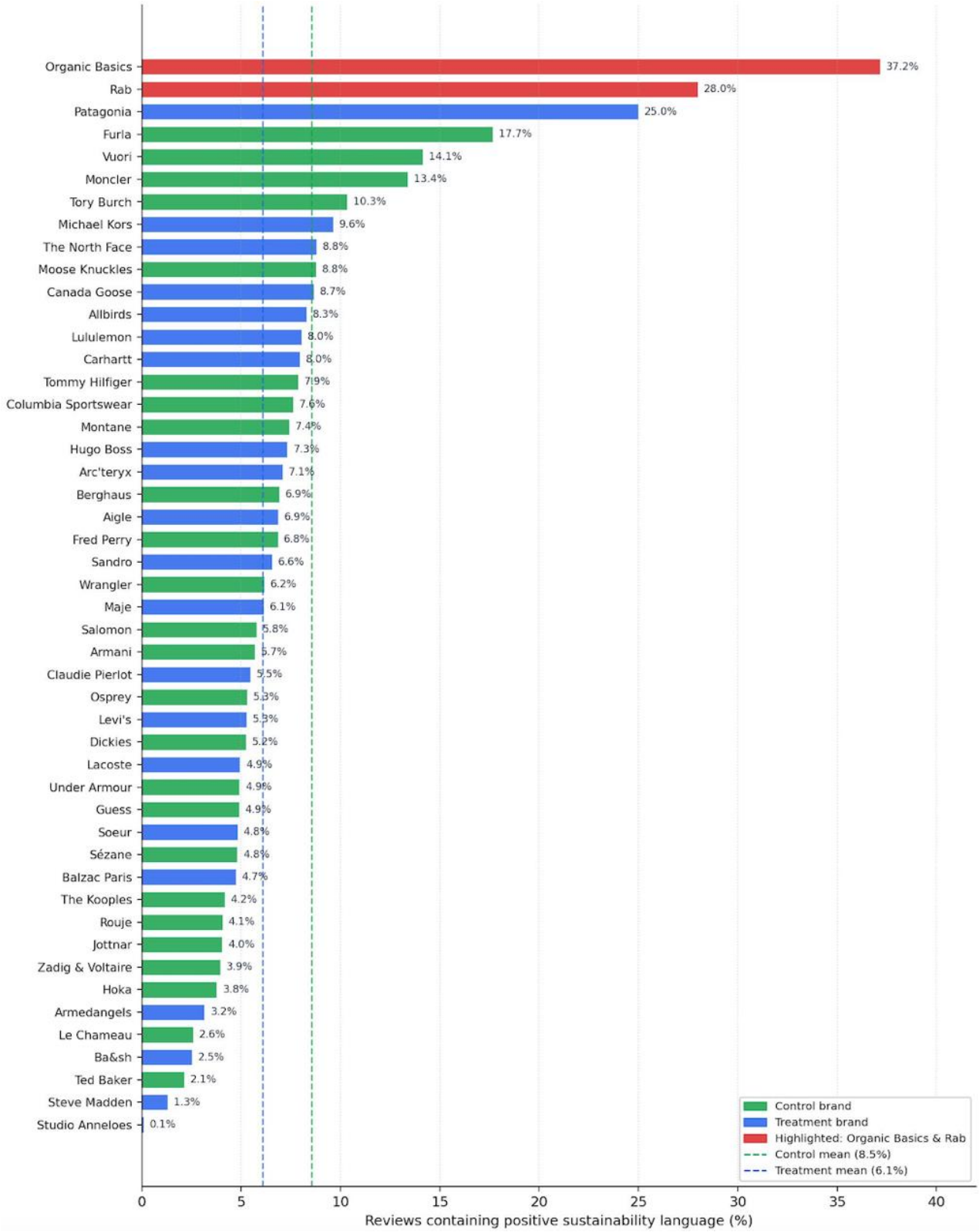
Topic	Label	Treatment Pre-launch	Treatment Post-launch	Control	Control Post-event
1	Service Complaints	42.8%	21.7%	59.2%	24.9%
2	Product Quality & Durability	25.4%	15.3%	18.0%	16.9%
3	Delivery & Fit	10.0%	18.5%	7.5%	35.5%
4	Service & Experience	4.7%	7.5%	9.3%	15.3%
5	Dutch-language reviews	4.0%	17.3%	0.1%	0.7%
6	French-language reviews	5.1%	12.6%	0.9%	3.9%
7	German-language reviews	8.0%	7.0%	5.1%	2.8%
	Total	100%	100%	100%	100%

Note: Proportions based on dominant-topic assignment. N = 5,456 pre-launch treatment reviews, 21,367 post-launch treatment reviews, 3,798 control pre-event reviews, 19,080 control post-event reviews (49,701 total after filtering reviews with fewer than 5 words). Control pre/post split uses pseudo-event dates inherited from matched treatment brands. The rise in Dutch and French language cluster shares post-launch reflects the differential growth of Studio Anneloes and French contemporary fashion brands in their post-launch periods, not a thematic shift in review content. The post-event rise in Delivery and Fit for control brands reflects the same dynamic: Studio Anneloes accounts for the bulk of control post-event reviews and its customers write heavily about delivery

A.5 Sustainability Language Mention

This metric counts the share of reviews that contain more positive sustainability words than sceptical ones, terms like recycling, circularity, or conscious consumption. Brands typically position resale as a sustainability initiative, so it matters whether customers actually notice or care about that framing. That is the question this measure tries to get at.

Figure A.5. Sustainability Language Mention Rate by Brand


Organic Basics 37.2%
Rab 28.0%
Patagonia 25.0%
Furla 17.7%
Vuori 14.1%
Moncler 13.4%
Tory Burch 10.3%
Michael Kors 9.6%
The North Face 8.8%
Moose Knuckles 8.8%
Canada Goose 8.7%
Allbirds 8.3%
Lululemon 8.0%
Carhartt 8.0%
Tommy Hilfiger 7.9%
Columbia Sportswear 7.6%
Montane 7.4%
Hugo Boss 7.3%
Arc'teryx 7.1%
Berghaus 6.9%
Aigle 6.9%
Fred Perry 6.8%
Sandro 6.6%
Wrangler 6.2%
Maje 6.1%
Salomon 5.8%
Armani 5.7%
Claudie Pierlot 5.5%
Osprey 5.3%
Levi's 5.3%
Dickies 5.2%
Lacoste 4.9%
Under Armour 4.9%
Guess 4.9%
Soeur 4.8%
Sézane 4.8%
Balzac Paris 4.7%
The Kooples 4.2%
Rouje 4.1%
Jottnar 4.0%
Zadig & Voltaire 3.9%
Hoka 3.8%
Armedangels 3.2%
Le Chameau 2.6%
Ba&sh 2.5%
Ted Baker 2.1%
Steve Madden 1.3%
Studio Anneloes 0.1%
Control brand
Treatment brand
Highlighted: Organic Basics & Rab
Control mean (8.5%)
Treatment mean (6.1%)
0
5
10
15
20
25
30
35
40
Reviews containing positive sustainability language (%)

A.6 Full Two-Way Fixed-Effects DiD Estimates

This table reports the full two-way fixed effects specification for all three outcomes, star rating, sustainability score, and value VADER, including standard errors, confidence intervals, within-R^2, and clustering details. The headline coefficients are summarized in the main text as part of the diagnostic comparison motivating the synthetic difference-in-differences approach; this table provides the complete model output for transparency.

Table A.6. Study 1: Two-Way Fixed-Effects DiD Estimates

	Star Rating	Sustainability score	Value VADER
β (DiD)	0.244	0.005	0.022
SE	0.233	0.021	0.054
t	1.045	0.227	0.415
p	0.301	0.822	0.680
95% CI	[−0.214, 0.701]	[−0.037, 0.046]	[−0.083, 0.128]
N	45,368	45,368	45,368
Within R^2	*0.441*	*0.087*	*0.225*
Brand FE	✓	✓	✓
Quarter-year FE	✓	✓	✓
SE clustering	*Brand*	*Brand*	*Brand*
Clusters	*46*	*46*	*46*

Note: β = DiD estimator (treated × post). SE = standard error clustered at the brand level. Within R^2 reported after brand and quarter-year demeaning. All models include brand and quarter-year fixed effects. N = 45,368 reviews across 46 brands (19 treatment, 27 control).

A.7 Robustness Check

Table A.7 compares the main synthetic DiD specification, using all 27 control brands, against a version that excludes the four low-volume brands from the donor pool. The pooled ATTs are comparable in direction and, if anything, slightly stronger without them: rating ATT = -0.167 (p = 0.079) versus -0.078 in the main specification, sustainability ATT = -0.031 (p =

0.003) versus -0.018, and value VADER ATT = -0.111 (p = 0.002) versus -0.103. The main results are not driven by the inclusion of the noisier lower-volume control brands.

Table A.7. Study 1: Robustness Check: Excluding Low-Volume Control Brands

Outcome	Main specification (27 controls) ATT (p)	Excluding low-volume controls (23 controls) ATT (p)
Star rating	-0.078 (0.419)	-0.167 (0.079)
Sustainability score	-0.018 (0.052)	-0.031 (0.003)
Value VADER	-0.103 (0.002)	-0.111 (0.002)

Note: Low-volume controls excluded are Tommy Hilfiger, Fred Perry, Jottnar, and Vuori (each under 150 total reviews). Synthetic DiD re-estimated using the remaining 23 control brands as the donor pool; all 19 treatment brands included as in the main specification.

A.8 Sensitivity Check: Moderation Analysis Using Raw Change Scores

Tables A.8a and A.8b repeat the moderation analysis using each brand's raw pre-to-post as the outcome instead of its synthetic DiD treatment effect, the naive comparison the main analysis was designed to correct for. The pattern differs from the main analysis: here, messaging intensity is significantly associated with value VADER (p = 0.026), and the integration × messaging interaction is significant for star rating (p = 0.020) rather than sustainability score. Because Δ does not account for what would have happened to each brand without the resale launch, this divergence is consistent with the main text's argument that the raw outcome is not causally informative on its own.

Table A.8.a Study 2: OLS Moderation Regressions (HC1 Robust Standard Errors)

Variable	Δ Rating			Δ Sust. Score			Δ VADER		
	β	**SE**	**p**	**β**	**SE**	**p**	**β**	**SE**	**p**
Integration level	−0.155	0.374	0.678	−0.033	0.035	0.345	−0.029	0.098	0.764
Messaging intensity	−0.362	0.211	0.085	0.011	0.015	0.458	−0.117	0.053	0.026 *
Discount depth (%)	−0.067	0.031	0.051	-0.001	0.003	0.731	−0.013	0.007	0.093
Service offerings	0.284	0.425	0.504	−0.043	0.037	0.249	0.082	0.095	0.389

Variable	Δ Rating			Δ Sust. Score			Δ VADER		
	β	SE	p	β	SE	p	β	SE	p
N	18			18			18		
R²	0.541			0.363			0.588		
Adjusted R²	0.290			0.016			0.363		

*Note: β = OLS coefficient; SE = HC1 heteroskedasticity-consistent standard error. Segment control dummies included in all models but not shown. * p < 0.05.*

Table A.8b. Study 2. Interaction Model: Δ Rating ~ Integration × Messaging

Variable	β	SE	t	p	95% CI
Integration level	1.644	0.721	2.282	0.023 *	[0.232, 3.057]
Messaging intensity	0.924	0.613	1.507	0.132	[−0.278, 2.125]
Integration × Messaging	−0.544	0.235	−2.319	0.020 *	[−1.004, −0.084]
Discount depth (%)	−0.061	0.033	−1.877	0.090	[−0.132, 0.010]
Service offerings	0.116	0.468	0.248	0.804	[−0.800, 1.032]
N	18				
R²	0.607				
Adjusted R²	0.333				

*Note: HC1 robust standard errors. Segment controls included but not shown. * p < 0.05.*